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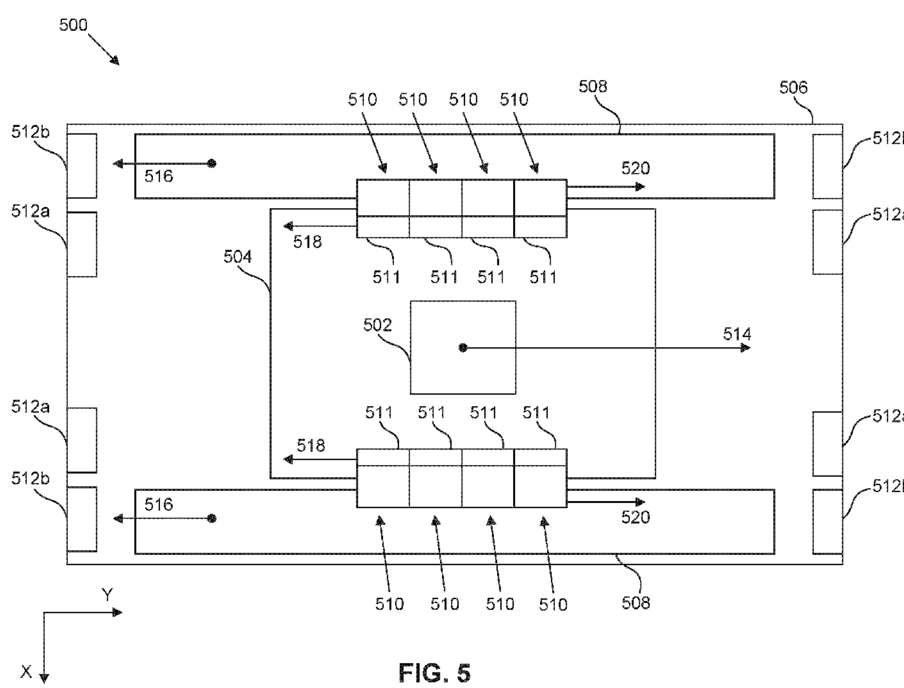

FIG. 5

(57) Abstract: A braking system can include a moving frame, one or more balance masses, one or more actuators, and one or more mechanical buffers. The moving frame can move with a predetermined kinetic energy. The one or more balance masses can absorb reaction forces exerted by the moving frame. The one or more actuators can move the moving frame and stop movement of the moving frame through braking to prevent collision damage in an error scenario. The one or more actuators can be electrically shorted to generate an electromotive braking force opposite to a direction of motion of the moving frame to reduce the kinetic energy of the moving frame and to reduce a braking distance of the moving frame. The one or more mechanical buffers can absorb a remainder of the kinetic energy of the moving frame remaining after the one or more actuators have completed their braking action.

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ELECTROMOTIVE FORCE BRAKING IN A LITHOGRAPHIC APPARATUS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority of US application 63/580,237 which was filed on September 1, 2023 and which is incorporated herein in its entirety by reference.

FIELD

[0002] The present disclosure relates to actuated stages, for example, a stage for supporting a reticle used in lithographic apparatuses and systems.

BACKGROUND

[0003] A lithographic apparatus is a machine constructed to apply a desired pattern onto a substrate. A lithographic apparatus can be used, for example, in the manufacture of integrated circuits (ICs). A lithographic apparatus can, for example, project a pattern of a patterning device (e.g., a mask, a reticle) onto a layer of radiation-sensitive material (photoresist or simply "resist") provided on a substrate.

[0004] To project a pattern on a substrate a lithographic apparatus can use electromagnetic radiation. The wavelength of this radiation determines the minimum size of features which can be formed on the substrate. A lithographic apparatus, which uses extreme ultraviolet (EUV) radiation, having a wavelength within the range 4-20 nm, for example 6.7 nm or 13.5 nm, can be used to form smaller features on a substrate than a lithographic apparatus which uses, for example, radiation with a wavelength of 193 nm.

[0005] A lithographic system can output only a finite number of fabricated devices in a given timeframe. Fast scanning of wafer stages and reticle stages can improve the speed of fabrication. However, efforts to increase velocities of movable stages can cause an increase in braking distance of the movable stages. Such an increase in braking distance can cause collisions resulting in mechanical damage in existing lithographic systems. For example, current mechanical buffers may not be able to absorb the kinetic energy of the movable stages independently, thereby failing to prevent the movable stages from causing damage to other components within the limited volume of an existing lithographic system.

SUMMARY

[0006] Accordingly, it is desirable to improve braking functions to be compatible with increased lithographic fabrication speed and throughput. Electromagnetic actuators for wafer and reticle stages can provide electromotive braking forces according to aspects described herein.

[0007] In some aspects, a lithographic apparatus can include an illumination system, a patterning system, a projection system, and a reticle stage. The illumination system can be configured to produce a beam of radiation. The patterning system can include a reticle and can be configured to impart a pattern on the beam. The projection system can be configured to project a patterned beam on a substrate. The reticle stage can include a moving frame, one or more balance masses, one or more actuators, and one or more mechanical buffers. The moving frame can move with a predetermined amount of kinetic energy. The one or more balance masses can absorb reaction forces exerted by the moving frame. The one or more actuators can move the moving frame and can stop movement of the moving frame through braking in response to an error scenario to prevent collision damage. The one or more actuators can be electrically shorted to generate an electromotive braking force opposite to a direction of motion of the moving frame to reduce the kinetic energy of the moving frame and to reduce a braking distance of the moving frame. The one or more mechanical buffers can absorb a remainder of the kinetic energy of the moving frame remaining after the one or more actuators have completed their braking action.

[0008] In some aspects, a braking system can include a moving frame, one or more balance masses, one or more actuators, and one or more mechanical buffers. The moving frame can move with a predetermined amount of kinetic energy. The one or more balance masses can absorb reaction forces exerted by the moving frame. The one or more actuators can move the moving frame and can stop movement of the moving frame through braking in response to an error scenario to prevent collision damage. The one or more actuators can be electrically shorted to generate an electromotive braking force opposite to a direction of motion of the moving frame to reduce the kinetic energy of the moving frame and to reduce a braking distance of the moving frame. The one or more mechanical buffers can absorb a remainder of the kinetic energy of the moving frame remaining after the one or more actuators have completed their braking action.

[0009] In some aspects, a braking method can include applying a current to one or more actuator coils to interact with a magnetic field from one or more magnets adjacent to the one or more actuator coils. The braking method can further include generating a force to move a moving frame in a lithographic apparatus in a first direction. The braking method can further include shorting the one or more actuator coils during a movement of the moving frame in response to an error scenario to prevent collision damage. The braking method can further include inducing currents in the one or more actuator coils caused by the magnetic field from the one or more magnets. The braking method can further include generating a magnetic field in the one or more actuator coils opposite to the magnetic field from the one or more magnets to produce an electromotive braking force in a second direction, opposite to the first direction of the movement of the moving frame. The braking method can further include reducing a velocity of the moving frame before the moving frame crashes into one or more mechanical buffers.

[0010] Further features of various aspects of the present disclosure are described in detail below with reference to the accompanying drawings. It is noted that the present disclosure is not limited to the specific aspects described herein. Such aspects are presented herein for illustrative purposes only. Additional aspects will be apparent to those skilled in the relevant art(s) based on the teachings contained herein.

BRIEF DESCRIPTION OF THE DRAWINGS/FIGURES

[0011] The accompanying drawings, which are incorporated herein and form part of the specification, illustrate the present disclosure and, together with the description, further serve to explain the principles of the present disclosure and to enable those skilled in the relevant art(s) to make and use aspects described herein.

[0012] FIG. 1A shows a reflective lithographic apparatus, according to some aspects.

[0013] FIG. 1B shows a transmissive lithographic apparatus, according to some aspects.

[0014] FIG. 1C shows a lithographic cell, according to some aspects.

[0015] FIGS. 2 and 3 show a reticle stage, according to some aspects.

[0016] FIG. 4 shows a reticle exchange apparatus, according to some aspects.

[0017] FIG. 5 shows an actuated stage, according to some aspects.

[0018] FIGS. 6A and 6B show actuator devices, according to some aspects.

[0019] FIGS. 7A and 7B show mechanical buffers, according to some aspects.

[0020] FIG. 8 shows a flowchart for a braking method, according to some aspects.

[0021] The features of the present disclosure will become more apparent from the detailed description set forth below when taken in conjunction with the drawings, in which like reference characters identify corresponding elements throughout. In the drawings, like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements. Additionally, generally, the left-most digit(s) of a reference number identifies the drawing in which the reference number first appears. Unless otherwise indicated, the drawings provided throughout the disclosure should not be interpreted as to-scale drawings.

DETAILED DESCRIPTION

[0022] The aspects described herein, and references in the specification to “one aspect,” “an aspect,” “an exemplary aspect,” “an example aspect,” etc., indicate that the aspects described can include a particular feature, structure, or characteristic, but every aspect may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same aspect. Further, when a particular feature, structure, or characteristic is described in connection with an aspect, it is understood that it is within the knowledge of those skilled in the art to effect such feature, structure, or characteristic in connection with other aspects whether or not explicitly described.

[0023] Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “on,” “upper” and the like, can be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus can be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein can likewise be interpreted accordingly.

[0024] The terms "about," "approximately," or the like can be used herein to indicate the value of a given quantity that can vary based on a particular technology. Based on the particular technology, the terms "about," "approximately," or the like can indicate a value of a given quantity that varies within, for example, 10–30% of the value (e.g., ±10%, ±20%, or ±30% of the value).

[0025] Aspects of the present disclosure can be implemented in hardware, firmware, software, or any combination thereof. Aspects of the disclosure can also be implemented as instructions stored on a computer-readable medium, which can be read and executed by one or more processors. A machine-readable medium can include any mechanism for storing or transmitting information in a form readable by a machine (e.g., a computing device). For example, a machine-readable medium can include read only memory (ROM); random access memory (RAM); magnetic disk storage media; optical storage media; flash memory devices; electrical, optical, acoustical or other forms of propagated signals (e.g., carrier waves, infrared signals, digital signals, etc.), and others. Furthermore, firmware, software, routines, and/or instructions can be described herein as performing certain actions. However, it should be appreciated that such descriptions are merely for convenience and that such actions result from computing devices, processors, controllers, or other devices executing the firmware, software, routines, instructions, etc. The term "machine-readable medium" can be interchangeable with similar terms, for example, "computer program product," "computer-readable medium," "non-transitory computer-readable medium," or the like. The term "non-transitory" can be used herein to characterize one or more forms of computer readable media except for a transitory, propagating signal.

[0026] Before describing such aspects in more detail, however, it is instructive to present an example environment in which aspects of the present disclosure can be implemented.

[0027] *Example Lithographic Systems*

[0028] FIGS. 1A and 1B show a lithographic apparatus 100 and a lithographic apparatus 100', respectively, in which aspects of the present disclosure can be implemented. Lithographic apparatus 100 and lithographic apparatus 100' each include the following: an illumination system (illuminator) IL configured to condition a radiation beam B (for example, deep ultra violet or extreme ultra violet radiation); a support structure (for example, a mask table) MT configured to support a patterning device (for example, a mask, a reticle, or a dynamic patterning device) MA and connected to a first positioner PM configured to accurately position the patterning device MA; and, a substrate table (for example, a wafer table) WT configured to hold a substrate (for example, a resist coated wafer) W and connected to a second positioner PW configured to accurately position the substrate W. Lithographic apparatus 100 and 100' also have a projection system PS configured to project a pattern imparted to the radiation beam B by patterning device MA onto a target portion (for example, comprising one or more dies) C of the substrate W. In lithographic apparatus 100, the patterning device MA and the projection system PS are reflective. In lithographic apparatus 100', the patterning device MA and the projection system PS are transmissive.

[0029] The illumination system IL can include various types of optical components, such as refractive, reflective, catadioptric, magnetic, electromagnetic, electrostatic, or other types of optical components, or any combination thereof, for directing, shaping, or controlling the radiation beam B.

[0030] The support structure MT holds the patterning device MA in a manner that depends on the orientation of the patterning device MA with respect to a reference frame, the design of at least one of the lithographic apparatus 100 and 100', and other conditions, such as whether or not the patterning device MA is held in a vacuum environment. The support structure MT can use mechanical, vacuum, electrostatic, or other clamping techniques to hold the patterning device MA. The support structure MT can be a frame or a table, for example, which can be fixed or movable. By using sensors, the support structure MT can ensure that the patterning device MA is at a desired position, for example, with respect to the projection system PS.

[0031] The term "patterning device" MA should be broadly interpreted as referring to any device that can be used to impart a radiation beam B with a pattern in its cross-section, such as to create a pattern in the target portion C of the substrate W. The pattern imparted to the radiation beam B can correspond to a particular functional layer in a device being created in the target portion C to form an integrated circuit.

[0032] The patterning device MA can be transmissive (as in lithographic apparatus 100' of FIG. 1B) or reflective (as in lithographic apparatus 100 of FIG. 1A). Examples of patterning devices MA include reticles, masks, programmable mirror arrays, or programmable LCD panels. Masks are well known in lithography, and include mask types such as binary, alternating phase shift, or attenuated phase shift, as well as various hybrid mask types. An example of a programmable mirror array employs a matrix arrangement of small mirrors, each of which can be individually tilted so as to reflect an incoming radiation beam in different directions. The tilted mirrors impart a pattern in the radiation beam B, which is reflected by a matrix of small mirrors.

[0033] The term "projection system" PS can encompass any type of projection system, including refractive, reflective, catadioptric, magnetic, electromagnetic and electrostatic optical systems, or any combination thereof, as appropriate for the exposure radiation being used, or for other factors, such as the use of an immersion liquid on the substrate W or the use of a vacuum. A vacuum environment can be used for EUV or electron beam radiation since other gases can absorb too much radiation or electrons. A vacuum environment can therefore be provided to the whole beam path with the aid of a vacuum wall and vacuum pumps.

[0034] Lithographic apparatus 100 and/or lithographic apparatus 100' can be of a type having two (dual stage) or more substrate tables WT (and/or two or more mask tables). In such "multiple stage" machines, the additional substrate tables WT can be used in parallel, or preparatory steps can be carried out on one or more tables while one or more other substrate tables WT are being used for exposure. In some situations, the additional table may not be a substrate table WT.

[0035] The lithographic apparatus can also be of a type wherein at least a portion of the substrate can be covered by a liquid having a relatively high refractive index, e.g., water, so as to fill a space between the projection system and the substrate. An immersion liquid can also be applied to other spaces in the lithographic apparatus, for example, between the mask and the projection system. Immersion techniques are well known in the art for increasing the numerical aperture of projection systems. The term "immersion" as used herein does not mean that a structure, such as a substrate, must be submerged in liquid. For example, a liquid can be located between the projection system and the substrate during exposure.

[0036] Referring to FIGS. 1A and 1B, the illuminator IL receives a radiation beam from a radiation source SO. The source SO and the lithographic apparatus 100, 100' can be separate physical entities, for example, when the source SO is an excimer laser. In such cases, the source SO is not considered to form part of the lithographic apparatus 100 or 100', and the radiation beam B passes from the source SO to the illuminator IL with the aid of a beam delivery system BD (in FIG. 1B) including, for example, suitable directing mirrors and/or a beam expander. In other cases, the source SO can be an integral part of the lithographic apparatus 100, 100', for example, when the source SO is a mercury lamp. A radiation system can comprise the source SO, the illuminator IL, and/or the beam delivery system BD.

[0037] The illuminator IL can include an adjuster AD (in FIG. 1B) for adjusting the angular intensity distribution of the radiation beam. Generally, at least the outer and/or inner radial extent (commonly referred to as "σ-outer" and "σ-inner," respectively) of the intensity distribution in a pupil plane of the illuminator can be adjusted. In addition, the illuminator IL can comprise various other components (in FIG. 1B), such as an integrator IN and a condenser CO. The illuminator IL can be used to condition the radiation beam B to have a desired uniformity and intensity distribution in its cross section.

[0038] Referring to FIG. 1A, the radiation beam B is incident on the patterning device (for example, mask) MA, which is held on the support structure (for example, mask table) MT, and is patterned by the patterning device MA. In lithographic apparatus 100, the radiation beam B is reflected from the patterning device (for example, mask) MA. After being reflected from the patterning device (for example, mask) MA, the radiation beam B passes through the projection system PS, which focuses the radiation beam B onto a target portion C of the substrate W. With the aid of the second positioner PW and position sensor IF2 (for example, an interferometric device, linear encoder, or capacitive sensor), the substrate table WT can be moved accurately (for example, so as to position different target portions C in the path of the radiation beam B). Similarly, the first positioner PM and another position sensor IF1 can be used to accurately position the patterning device (for example, mask) MA with respect to the path of the radiation beam B. Patterning device (for example, mask) MA and substrate W can be aligned using mask alignment marks M1, M2 and substrate alignment marks P1, P2.

[0039] Referring to FIG. 1B, the radiation beam B is incident on the patterning device (for example, mask MA), which is held on the support structure (for example, mask table MT), and is patterned by the patterning device. Having traversed the mask MA, the radiation beam B passes through the projection system PS, which focuses the beam onto a target portion C of the substrate W. The projection system has a pupil conjugate PPU to an illumination system pupil IPU. Portions of radiation emanate from the intensity distribution at the illumination system pupil IPU and traverse a mask pattern without being affected by diffraction at the mask pattern and create an image of the intensity distribution at the illumination system pupil IPU.

[0040] The projection system PS projects an image of the mask pattern MP, where the image is formed by diffracted beams produced from the mark pattern MP by radiation from the intensity distribution, onto a photoresist layer coated on the substrate W. For example, the mask pattern MP can include an array of lines and spaces. A diffraction of radiation at the array and different from zeroth order diffraction generates diverted diffracted beams with a change of direction in a direction perpendicular to the lines. Undiffracted beams (i.e., so-called zeroth order diffracted beams) traverse the pattern without any change in propagation direction. The zeroth order diffracted beams traverse an upper lens or upper lens group of the projection system PS, upstream of the pupil conjugate PPU of the projection system PS, to reach the pupil conjugate PPU. The portion of the intensity distribution in the plane of the pupil conjugate PPU and associated with the zeroth order diffracted beams is an image of the intensity distribution in the illumination system pupil IPU of the illumination system IL. The aperture device PD, for example, is disposed at or substantially at a plane that includes the pupil conjugate PPU of the projection system PS.

[0041] The projection system PS is arranged to capture (e.g., using a lens or lens group L) the zeroth order diffracted beams, first order diffracted beams, and/or higher order diffracted beams (not shown). In some aspects, dipole illumination for imaging line patterns extending in a direction perpendicular to a line can be used to utilize the resolution enhancement effect of dipole illumination. For example, first-order diffracted beams interfere with corresponding zeroth-order diffracted beams at the level of the wafer W to create an image of the line pattern MP at highest possible resolution and process window (i.e., usable depth of focus in combination with tolerable exposure dose deviations). In some aspects, astigmatism aberration can be reduced by providing radiation poles (not shown) in opposite quadrants of the illumination system pupil IPU. Further, in some aspects, astigmatism aberration can be reduced by blocking the zeroth order beams in the pupil conjugate PPU of the projection system associated with radiation poles in opposite quadrants. This is described in more detail in US 7,511,799 B2, issued Mar. 31, 2009, which is incorporated by reference herein in its entirety.

[0042] With the aid of the second positioner PW and position sensor IFD (for example, an interferometric device, linear encoder, or capacitive sensor), the substrate table WT can be moved accurately (for example, so as to position different target portions C in the path of the radiation beam B). Similarly, the first positioner PM and another position sensor (not shown in FIG. 1B) can be used to accurately position the mask MA with respect to the path of the radiation beam B (for example, after mechanical retrieval from a mask library or during a scan).

[0043] In general, movement of the mask table MT can be realized with the aid of a long-stroke module (coarse positioning) and a short-stroke module (fine positioning), which form part of the first positioner PM. Similarly, movement of the substrate table WT can be realized using a long-stroke module and a short-stroke module, which form part of the second positioner PW. In the case of a stepper (as opposed to a scanner), the mask table MT can be connected to a short-stroke actuator or can be fixed. Mask MA and substrate W can be aligned using mask alignment marks M1, M2, and substrate alignment marks P1, P2. Although the substrate alignment marks (as illustrated) occupy dedicated target portions, they can be located in spaces between target portions (known as scribe-lane alignment marks). Similarly, in situations in which more than one die is provided on the mask MA, the mask alignment marks can be located between the dies.

[0044] Mask table MT and patterning device MA can be in a vacuum chamber V, where an in-vacuum robot IVR can be used to move patterning devices such as a mask in and out of vacuum chamber. Alternatively, when mask table MT and patterning device MA are outside of the vacuum chamber, an out-of-vacuum robot can be used for various transportation operations, similar to the in-vacuum robot IVR. Both the in-vacuum and out-of-vacuum robots can be calibrated for a smooth transfer of any payload (e.g., mask) to a fixed kinematic mount of a transfer station.

[0045] The lithographic apparatus 100 and 100' can be used in at least one of the following modes:

1. In step mode, the support structure (for example, mask table) MT and the substrate table WT are kept essentially stationary, while an entire pattern imparted to the radiation beam B is projected onto a target portion C at one time (i.e., a single static exposure). The substrate table WT is then shifted in the X and/or Y direction so that a different target portion C can be exposed.

2. In scan mode, the support structure (for example, mask table) MT and the substrate table WT are scanned synchronously while a pattern imparted to the radiation beam B is projected onto a target portion C (i.e., a single dynamic exposure). The velocity and direction of the substrate table WT relative to the support structure (for example, mask table) MT can be determined by the (de-)magnification and image reversal characteristics of the projection system PS.

3. In another mode, the support structure (for example, mask table) MT is kept substantially stationary holding a programmable patterning device, and the substrate table WT is moved or scanned while a pattern imparted to the radiation beam B is projected onto a target portion C. A pulsed radiation source SO can be employed and the programmable patterning device is updated as needed after each movement of the substrate table WT or in between successive radiation pulses during a scan. This mode of operation can be readily applied to maskless lithography that utilizes a programmable patterning device, such as a programmable mirror array.

[0046] Combinations and/or variations on the described modes of use or entirely different modes of use can also be employed.

[0047] In some aspects, lithographic apparatus 100 includes an extreme ultraviolet (EUV) source, which is configured to generate a beam of EUV radiation for EUV lithography. In general, the EUV source is configured in a radiation system, and a corresponding illumination system is configured to condition the EUV radiation beam of the EUV source.

[0048] In some aspects, lithographic apparatus 100' includes a deep ultraviolet (DUV) source, which is configured to generate a beam of DUV radiation for DUV lithography. In general, the DUV source is configured in a radiation system, and a corresponding illumination system is configured to condition the DUV radiation beam of the DUV source.

[0049] *Example Lithographic Cell*

[0050] FIG. 1C shows a lithographic cell 102, also sometimes referred to a lithocell or cluster, according to some aspects. Lithographic apparatuses 100 or 100' can form part of lithographic cell 100. Lithographic cell 102 can also include one or more apparatuses to perform pre- and post-exposure processes on a substrate. Conventionally these include spin coaters SC to deposit resist layers, developers DE to develop exposed resist, chill plates CH, and bake plates BK. A substrate handler, or robot, RO picks up substrates from input/output ports I/O1, I/O2, moves them between the different process apparatuses and delivers them to the loading bay LB of the lithographic apparatus 100 or 100'. These devices, which are often collectively referred to as the track, are under the control of a track control unit TCU, which is itself controlled by a supervisory control system SCS, which also controls the lithographic apparatus via lithography control unit LACU. Thus, the different apparatuses can be operated to maximize throughput and processing efficiency.

[0051] *Example Reticle Stage*

[0052] FIGS. 2 and 3 show a reticle stage 200, according to some aspects. Reticle stage 200 can include top stage surface 202, bottom stage surface 204, side stage surfaces 206, and clamp 300. In some aspects, reticle stage 200 with clamp 300 can be implemented in lithographic apparatus LA. For example, reticle stage 200 can be support structure MT in lithographic apparatus LA. In some aspects, clamp 300 can be disposed on top stage surface 202. For example, as shown in FIG. 2, clamp 300 can be disposed at a center of top stage surface 202 with clamp frontside 302 facing perpendicularly away from top stage surface 202.

[0053] In some lithographic apparatuses, for example, lithographic apparatus LA, a reticle stage 200 with a clamp 300 can be used to hold and position a reticle 408 for scanning or patterning operations. In one example, the reticle stage 200 can rely on powerful drives, large balance masses, and heavy frames to support it. In one example, the reticle stage 200 can have a large inertia and can weigh over 500 kg to propel and position a reticle 408 weighing about 0.5 kg. To accomplish reciprocating motions of the reticle 408, which are typically found in lithographic scanning or patterning operations, accelerating and decelerating forces can be provided by linear motors that drive the reticle stage 200.

[0054] In some aspects, as shown in FIGS. 2 and 3, reticle stage 200 can include first encoder 212 and second encoder 214 for positioning operations. For example, first and second encoders 212 and 214 can be interferometers. First encoder 212 can be attached along a first direction, for example, a transverse direction (i.e., X-direction) of reticle stage 200. And second encoder 214 can be attached along a second direction, for example, a longitudinal direction (i.e., Y-direction) of reticle stage 200. In some aspects, as shown in FIGS. 2 and 3, first encoder 212 can be orthogonal to second encoder 214.

[0055] As shown in FIGS. 2 and 3, reticle stage 200 can include clamp 300. Clamp 300 is configured to hold reticle 408 in a fixed plane on reticle stage 200. Clamp 300 includes clamp frontside 302 and can be disposed on top stage surface 202. In some aspects, clamp 300 can use mechanical, vacuum, electrostatic, or other suitable clamping techniques to hold and secure an object. In some aspects, clamp 300 can be an electrostatic clamp, which can be configured to electrostatically clamp (i.e., hold) an object, for example, reticle 408 (FIG. 4) in a vacuum environment. For EUV generation performed in a vacuum environment, it can be difficult to use vacuum clamps to clamp a mask or reticle. Instead, electrostatic clamp(s) can be used. For example, clamp 300 can include an electrode, a resistive layer on the electrode, a dielectric layer on the resistive layer, and burls projecting from the dielectric layer. In use, a voltage can be applied to clamp 300, for example, several kV. And current can flow through the resistive layer, such that the voltage at an upper surface of the resistive layer will substantially be the same as the voltage of the electrode and generate an electric field. Also, a Coulomb force, attractive force between electrically opposite charged particles, will attract an object to clamp 300 and hold the object in place. In some aspects, clamp 300 can be a rigid material, for example, a metal, a dielectric, a ceramic, or a combination thereof.

[0056] *Example Reticle Exchange Apparatus*

[0057] FIG. 4 shows a reticle exchange apparatus 401, according to some aspects. Reticle exchange apparatus 401 can be configured to minimize reticle exchange time, particle generation, and contact forces or stresses from clamp 300 and/or reticle 408 to reduce damage to clamp 300 and reticle 408 and increase overall throughput in a reticle exchange process, for example, in a lithographic apparatus LA.

[0058] Reticle exchange apparatus 401 can include reticle stage 200, clamp 300, and in-vacuum robot 400. In-vacuum robot 400 can include reticle handler 402.

[0059] In some aspects, reticle handler 402 can be a rapid exchange device (RED), which is configured to efficiently rotate and minimize reticle exchange time. For example, reticle handler 402 can save time by moving multiple reticles from one position to another substantially simultaneously, instead of serially.

[0060] In some aspects, reticle handler 402 can include one or more reticle handler arms 404. Reticle handler arm 404 can include reticle baseplate 406. Reticle baseplate 406 can be configured to hold an object, for example, reticle 408.

[0061] In some aspects, reticle baseplate 406 can be an extreme ultraviolet inner pod (EIP) for a reticle. In some aspect, reticle baseplate 406 includes reticle baseplate frontside 407, and reticle 408 includes reticle backside 409.

[0062] In some aspects, reticle baseplate 406 can hold reticle 408 such that reticle baseplate frontside 407 and reticle backside 409 each face top stage surface 202 and clamp frontside 302. For example, reticle baseplate frontside 407 and reticle backside 409 can be facing perpendicularly away from top stage surface 202 and clamp frontside 302.

[0063] In some aspects, as shown in FIG. 4, reticle handler arms 404 can be arranged symmetrically about reticle handler 402. For example, reticle handler arms 404 can be spaced from each other by about 90 degrees, 120 degrees, or 180 degrees. In some aspects, reticle handler arms 404 can be arranged asymmetrically about reticle handler 402. For example, two reticle handler arms 404 can be spaced from each other by about 135 degrees, while another two reticle handler arms 404 can be spaced from each other by about 90 degrees.

[0064] In some aspects, during a reticle exchange process, reticle stage 200 with clamp 300 can be adjusted in a multi-stage movement (e.g., long stroke stage (coarse motion), short stroke stage (fine motion)).

[0065] *Example Actuated Stage*

[0066] Some aspects described herein provide structures and functions to address issues related to preventing mechanical collision damage in a lithographic apparatus.

[0067] FIG. 5 shows a stage 500 for supporting an object 502, according to some aspects. In some aspects, stage 500 can represent a different view of reticle stage 200 to emphasize additional details. Stage 500 can comprise an object 502, a moving frame 504, a base frame 506, balance masses 508, actuator devices 510, and mechanical buffers 512a, 512b.

[0068] In some aspects, stage 500 can be used in lithographic apparatus LA (FIG. 1), a lithographic cell (e.g., an arrangement of a multiple lithographic apparatuses), inspection apparatuses, or any apparatus in general that has a stage implementation for supporting and moving an object. For example, stage 500 can show a specific implementation of wafer table WT or mask table MT (FIG. 1).

[0069] In some aspects, moving frame 504 can be an actuated structure (e.g., for coarse motion of object 502). In a lithographic fabrication process, object 502 can be a reticle, a wafer, or the like. Furthermore, stage 500 can also include additional movement budget to shuttle object 502 to and from a loading area. Therefore, moving frame 504 can be responsible for a coarse motion of stage 500, e.g., in the order of tens, hundreds, or thousands of millimeters. Other distances may be chosen based on suitability for a particular implementation.

[0070] In some aspects, object 502 can be temporarily affixed onto moving frame 504 by pressing object 502 onto moving frame 504. This can be accomplished by vacuum clamping (suction force), electrostatic clamping (electrostatic force), mechanical clamping, or the like. Under ideal conditions, mutual friction between object 502 (e.g., a reticle) and moving frame 504 (e.g., a chuck) can ensure that there is no slippage therebetween. However, mechanical stresses due to high accelerations can induce some slippage, resulting in printing error. The errors can be highly detrimental due to the possibility of losing thousands of device products by the time the error can be detected.

[0071] In some aspects, moving frame 504 can be coupled to balance masses 508 adjacent to moving frame 504. In some aspects, balance masses 508 can be configured to mimic coarse movements, dampen, and return moving frame 504 to an equilibrium position. In some aspects, balance masses 508 can be configured to absorb reaction forces exerted by moving frame 504. In some aspects, balance masses 508 can be supported by base frame 506 while still allowing relative movement between moving frame 504 and balance masses 508. The motion of moving frame 504 can be limited to an axis (e.g., Y-axis) using guide rails or a contactless method (e.g., magnetic levitation) (guide devices not shown). The coordinate axes X and Y are provided as an example and are not to be construed as limiting.

[0072] In some aspects, actuator devices 510 can be responsible for course motion of moving frame 504. Actuator devices 510 can be arranged to accelerate moving frame 504 by providing a driving force between moving frame 504 and balance masses 508. The driving force accelerates moving frame 504 in a desired direction. Due to the conservation of momentum, the driving force is also applied to balance masses 508 with equal magnitude, but at a direction opposite to the desired direction. Typically, a mass of the balance masses 508 is significantly larger than a mass of the moving frame 504.

[0073] In some aspects, actuator devices 510 can comprise coils (e.g., wires that are coiled around a ferromagnetic core) and magnets. In some aspects, the magnets can comprise a material that responds to magnetic fields (e.g., a metal, iron, ferrite, or the like). In some aspects, actuator device 510 can include a mover 511 or 511' affixed to moving frame 504. Actuator devices 510 can actuate moving frame 504 by interacting the coils with the magnets to provide a driving force that moves mover 511 or 511'. In some aspects, mover 511 (shown in FIG. 6A) can include coils (e.g., coils 626a-626f shown in FIG. 6A) configured to interact with the magnets (e.g., magnets 622 shown in FIG. 6A) of actuator devices 510. In some aspects, mover 511' (shown in FIG. 6B) can include magnets (e.g., magnets 622 shown in FIG. 6B) configured to interact with the coils (e.g., coils 626a-626f shown in FIG. 6B) of actuator devices 510.

[0074] In some aspects, actuator devices 510 can include electromagnets configured to generate and adjust magnetic fields. An electromagnet can comprise coils (e.g, coils 626a-626f as shown in FIG. 6A and 6B) of wire wrapped around a metal core (e.g., a ferrite core). Actuator devices 510 can make the coils repel and attract the magnets by reversing a direction of the magnetic field. The actuator setup described herein can be referred to by other terms of art (e.g., a linear actuator). The number and configuration of actuator-related elements are not limited to those shown in FIG. 5. Fewer or more actuator-related elements can be used, as well as other configurations.

[0075] In some aspects, actuator devices 510 can actuate moving frame 504 using a high acceleration. The acceleration can be, for example, approximately 4–100g, 10–50g, 20–40g, or the like (where g is 9.8 m/s^2). A high acceleration can increase lithographic print production (e.g., increase throughput). Lithographic pattern transfer can be performed when moving frame 504 is in motion, for example, when it reaches a constant coasting speed. Coasting speeds can be, for example, 0.5–10.0 m/s, 1.0–7.0 m/s, 3.0–5.0 m/s, or the like. In some aspects, moving frame 504 can be configured to move with a predetermined amount of kinetic energy based on these coasting speeds. Performing the pattern transfer at a constant scanning speed can result in more accurate transfers of the printed pattern, whereas printing during acceleration can be accompanied by larger positional uncertainties.

[0076] The term "throughput" can be understood as the amount of material or items passing through a system or process. In some aspects, the term "throughput" can be used to characterize a rate of lithographic fabrication. For example, throughput can refer to a rate at which lithographic fabrication is completed on wafers, a rate at which a wafer clears a particular fabrication step and moves to the next step, or the like. Throughput can be a performance marker of a lithographic apparatus. It is desirable for lithographic systems to output as many products as possible in as little time as possible. Lithographic fabrication can comprise several complex processes. Each part of the process can involve tradeoffs that balance quality (e.g., sub-nanometer accuracy, high yield) and drawbacks (e.g., slower fabrication, cost). For example, to improve pattern-transfer speeds, lithography can implement faster, yet accurate, actuation of substrates and/or masks.

[0077] However, implementing faster actuation can put a lithographic apparatus at an increased risk of mechanical damage in the event of an error scenario. An error scenario can include failures involving actuators, software, temperature, etc. As a result, an error scenario can prompt a need to shut off actuators for material and machine damage control. Shutting off actuators can put a lithographic apparatus at risk of a collision event, in which a moving stage may crash into the base frame. Therefore, a lithographic apparatus can use devices to dissipate all kinetic energy of a moving stage. Dissipating the kinetic energy can be accomplished by electromotive force braking, traditional mechanical buffers, or both.

[0078] In some aspects, FIG. 5 shows motion of components of stage 500 when experiencing an error scenario. When an error scenario occurs, moving frame 504 can be moving with a first-direction velocity 514. Balance masses 508 can be force-coupled with moving frame 504, thereby resulting in second-direction velocity 516 for balance masses 508. In some aspects, actuator devices 510 can be configured to stop movement of moving frame 504 through braking in response to an error scenario to prevent collision damage.

[0079] To reduce a first-direction velocity 514 of moving frame 504, one or more actuator devices 510 can be electrically shorted. In some aspects, actuator devices 510 can be selectively shorted in real time. Electrically shorting the actuator devices 510 can induce a current in the one or more actuator devices 510 as the coils move relative to the magnets.

[0080] Inducing the current in the coils can generate a magnetic field in the one or more coils opposite to a magnetic field from the one or more magnets, thereby generating an electromotive braking force 518 in a second direction, opposite to a direction of motion of first-direction velocity 514 of moving frame 504. Electromotive braking force 518 can produce a reaction force 520 on balance masses 508 in a first direction, opposite to the second-direction velocity 516 of balance masses 508. Electromotive braking force 518 and reaction force 520 can reduce first-direction velocity 514 of moving frame 504 and second-direction velocity 516 of balance masses 508 before moving frame 504 crashes into mechanical buffers 512a and balance masses 508 crash into mechanical buffers 512b. Therefore, electromotive braking force 518 can reduce the kinetic energy of moving frame 504.

[0081] In some aspects, mechanical buffers 512a can be shock absorbers, dampers, springs, or a combination thereof used to absorb a remainder of the kinetic energy of the moving frame 504 remaining after the one or more actuator devices 510 have completed their braking action in response to the error scenario. In some aspects, mechanical buffers 512b can be shock absorbers, dampers, springs, or a combination thereof used to absorb a remainder of the kinetic energy of the balance masses 508 remaining after the one or more actuator devices 510 have completed their braking action in response to the error scenario. In some aspects, mechanical buffers 512a, 512b can be shock absorbers, dampers, springs, or a combination thereof.

[0082] In some aspects, electromotive force braking can reduce a braking distance of moving frame 504. Therefore, electromotive force braking can enable a reduction in size of mechanical buffers 512a, 512b for efficient use of a limited volume in a lithographic apparatus.

[0083] In some aspects, moving frame 504 can operate in a vacuum environment. Mechanical buffers using dampers in a vacuum environment may leak hydraulic fluid and contaminate the environment. As a result, such dampers should be used in a vacuum environment. Therefore, electromotive braking force 518 can provide a damping function in a vacuum environment without the drawbacks of a mechanical buffer damper.

[0084] In some aspects, electromotive force braking can be used to reduce a velocity of a wafer stage. In some aspects, moving frame 504 can be a wafer stage (e.g., WT shown in FIGS. 1A and 1B). In some aspects, moving frame 504 (e.g., WT shown in FIGS. 1A and 1B) can include one or more actuator devices 510 configured to be electrically shorted to generate an electromotive braking force 518 opposite to a first-direction velocity 514 of moving frame 504 (e.g., WT shown in FIGS. 1A and 1B) on a linear axis to reduce a braking distance of moving frame 504 (e.g., WT shown in FIGS. 1A and 1B) to prevent collision damage.

[0085] FIGS. 6A and 6B show a schematic cross-sectional illustration of actuator devices 510 and 510’, respectively, according to some aspects. It is to be appreciated that the description of FIGS. 6A and 6B can also refer to aspects of FIG. 5 when discussing a 5xx element. Although actuator devices 510 and 510’ are shown in FIGS. 6A and 6B as a stand-alone apparatus and/or system, the aspects of this disclosure can be used with other apparatuses, systems, and/or methods, such as, but not limited to, lithographic apparatus LA, support structure MT, and substrate table WT. In some aspects, actuator device 510 or 510’ can be part of a lithography apparatus, for example, lithographic apparatus LA. In some aspects, actuator device 510 or 510’ can be part of a support structure, for example, support structure MT for patterning device MA. In some aspects, actuator device 510 or 510’ can include magnets 622, back-iron plates 624, a mover 511 or 511’, coils 626a-626f, and a controller 628.

[0086] In some aspects, actuator device 510 or 510’ can be a linear actuator configured to provide a driving force along a single axis, for example the Y-axis. Multiple linear actuators can be applied to provide driving forces along multiple axes. In some aspects, actuator device 510 or 510' can be a planar actuator to provide a driving force along multiple axes. For example, the planar actuator can be arranged to move substrate table WT in 6 degrees of freedom.

[0087] In some aspects, actuator device 510 or 510' can be an electro-magnetic actuator comprising at least one coil and at least one magnet. Actuator device 510 or 510' can be arranged to move the at least one coil relative to the at least one magnet by applying an electrical current to the at least one coil. In some aspects, actuator device 510 can be a moving-coil type actuator which has the at least one coil coupled to the moving frame 504 and one or more coils move at high velocities adjacent to one or more stationary magnets. In some aspects, actuator device 510' can be a moving-magnet type actuator, which has the at least one magnet coupled to the moving frame 504 and one or more magnets move at high velocities adjacent to one or more stationary coils. In some aspects, actuator device 510 or 510' can be a voice-coil actuator, a reluctance actuator, a Lorentz-actuator or a piezo-actuator, or any other suitable actuator.

[0088] In some aspects, as shown in FIG. 6A, actuator device 510 can include an array of multiple permanent magnets 622 that are arranged in two parallel planes extending in an Y direction and an X direction perpendicular to the Y direction and a Z direction. Each array of permanent magnets 622 forming one of the two parallel planes can be supported by a respective back-iron plate 624. The back-iron plate 624 can support the magnets 622 and can keep a magnetic field generated by the magnets 622 within actuator device 510, thereby improving efficiency of actuator device 510. In some aspects, back-iron plates 624 can be affixed to balance masses 508 shown in FIG. 5.

[0089] In some aspects, as shown in FIG. 6A, a mover 511, moveable relative to the magnets 622, can be disposed between the two planes of permanent magnets 622. In some aspects, mover 511 can carry a coil arrangement comprising multiple coils 626a-626f, thereby forming a multiphase coil arrangement. Each coil 626a-626f can be arranged such that a current running through each coil 626a-626f is able to interact with magnetic fields produced by the permanent magnets 622 in order to generate Lorentz forces in a main direction, here the Y direction. The coil arrangement in mover 511 can include a first coil portion formed by coils 626a-626c and a second coil portion formed by coils 626d-626f. In the aspect as shown in FIG. 6A, the first coil portion formed by coils 626a-626c and a second coil portion formed by coils 626d-626f can be arranged to generate substantially similar Lorentz forces in the main direction when a substantially similar current is provided to the first coil portion formed by coils 626a-626c and to the second coil portion formed by coils 626d-626f. In some aspects, mover 511 can be affixed to moving frame 504 shown in FIG. 5 such that the generated Lorentz forces move moving frame 504 along a longitudinal axis (e.g., Y-axis).

[0090] In some aspects, the coil arrangement can be a three-phase coil arrangement, wherein coils 626a, 626d can form a first phase, coils 626b, 626e can form a second phase, and coils 626c, 626f can form a third phase. In some aspects, an AC current running through the first to third phases can be sinusoidal with a 120-degree phase shift between each phase. This aspect is merely one non-limiting example and other phase configurations can be implemented.

[0091] In some aspects, a controller 628 coupled to actuator device 510 can be configured to control movement (e.g., translation) of mover 511 along a longitudinal axis of stage 500 (e.g., Y-axis). In some aspects, controller 628 can be coupled (e.g., electronically) to coils 626a-626f. In some aspects, controller 628 can adjust a magnitude and a direction of currents running through coils 626a-626f to adjust a magnetic field strength and magnetic force. In some aspects, controller 628 can selectively short its corresponding actuator device 510 to redirect currents running through coils 626a-626f. For example, controller 628 can short actuator device 510 to induce an opposite current caused by relative motion between coils 626a-626f and magnets 622, thereby generating electromotive braking force 518.

[0092] In some aspects, moving frame 504 as shown in FIG. 5 can be moving with first-direction velocity 514 during an error scenario. To prevent mechanical damage caused by a collision, controller 628 can selectively short its corresponding actuator device 510 to redirect currents running through coils 626a-626f. In some aspects, controller 628 can short actuator device 510 to induce an opposite current running through coils 626a-626f caused by a relative motion between magnets 622 and coils 626a-626f. The induced opposite current running through coils 626a-626f can generate electromotive braking force 518 in a second direction opposite to first-direction velocity 514 of moving frame 504 shown in FIG. 5. In some aspects, electromotive braking force 518 can reduce first-direction velocity 514, and therefore a kinetic energy, of moving frame 504 shown in FIG. 5. In some aspects, controller 628 can be equipped with a capacitor configured to charge one or more actuators 510 with energy captured from the electromotive braking force 518.

[0093] In some aspects, electromotive braking force 518 can be defined by the following equation:

$$\begin{bmatrix} \dot{v} \\ \dot{F} \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{m} \\ -\frac{2}{3}\frac{K_m^2}{L} & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} v \\ F \end{bmatrix}$$

wherein $\dot{v}$ is the derivative of velocity with respect to time, $\dot{F}$ is the derivative of electromotive force with respect to time, v is the velocity, F is the electromotive force, m is mass of the moving frame, coil parameter K_m is a motor constant, coil parameter R is resistance, and coil parameter L is inductance. In some aspects, electromotive braking force 518 can depend on the above-mentioned coil parameters, mass, and velocity.

[0094] It will be apparent to the skilled artisan that althoug[h] ... n the aspect as shown in FIG. 6A is a moving coil configuration, an actuator can be a moving magnet ac... tuator in the aspect shown in FIG. 6B (e.g. actuator device 510'). In some aspects, multiple magnets 62... 2 can be affixed to each other to form mover 511', moveable between the two parallel planes of coils 6... 26a-626f. In some aspects, back-iron plates

624 can be affixed to balance masses 508 shown in FIG. 5. In some aspects, mover 511' can be affixed to moving frame 504 shown in FIG. 5 such that generated Lorentz forces move moving frame 504 along a longitudinal axis (e.g., Y-axis). In some aspects, the moving magnet configuration of actuator device 510' can include a coil arrangement with multiple coils (a first coil portion formed by coils 626a-626c and a second coil portion formed by coils 626d-626f) arranged in two parallel planes. In some aspects, actuator device 510' can be a multiphase coil arrangement, wherein each phase comprises two coils (coils 626a, 626d can form a first phase, coils 626b, 626e can form a second phase, and coils 626c, 626f can form a third phase). In some aspects, controller 628 coupled to actuator device 510' can be configured to control movement (e.g., translation) of mover 511' along a longitudinal axis of stage 500 (e.g., Y-axis). In some aspects, controller 628 can generate electromotive braking force 518 caused by a relative motion of magnets 622 and coils 626a-626f as described in relation to FIG. 6A.

[0095] FIGS. 7A and 7B show mechanical buffers, according to some aspects. It is to be appreciated that the description of FIGS. 7A and 7B can also refer to aspects of FIG. 5 when discussing a 5xx element. For example, in FIGS. 7A and 7B, mechanical buffer 512 or 512' can be an aspect of mechanical buffer 512a shown in FIG. 5 or an aspect of mechanical buffer 512b shown in FIG. 5. In FIG. 7A, a mechanical buffer 512 can include a damper 730, a spring 732, a guide pin 734, a collision site 736, and a mounting site 738. In FIG. 7B, a mechanical buffer 512' can include a spring 732, a guide pin 734, a collision site 736, and a mounting site 738.

[0096] In some aspects, mechanical buffers 512 or 512' can be affixed at mounting site 738 to base frame 506 shown in FIG. 5 to prevent collisions of fast moving objects (e.g., in the event of an error scenario in a lithographic apparatus). It can be undesirable to allow moving frame 504 or balance masses 508 to collide with base frame 506. Collisions can cause structures to become damaged and/or contaminate the clean lithographic environment by releasing contamination particles that can land on reticles and/or wafers. Therefore, moving frame 504, balance masses 508, and base frame 506 can be protected by mechanical buffers 512 or 512'. In some aspects, mechanical buffers 512 or 512' can be affixed at mounting site 738 to base frame 506 using any suitable method (e.g., adhesively bonded, bolted, nailed, clamped, or the like). In some aspects, mechanical buffers 512 or 512' can be arranged to maintain a non-zero gap distance between moving frame 504 and base frame 506. In some aspects, mechanical buffers 512 or 512' can be arranged to maintain a non-zero gap distance between balance masses 508 and base frame 506.

[0097] In some aspects, a mechanical buffer can comprise any collapsible structure (e.g., springs, flexures, hydraulics, or the like).

[0098] In some aspects, a guide pin 734 can be disposed inside a spring 732 to prevent undesirable deformation during collapse.

[0099] In one exemplary aspect, mechanical buffer 512 can include a damper 730, configured to absorb kinetic energy from moving frame 504 or balance masses 508, and a spring 732, configured to push moving frame 504 or balance masses 508 in an opposite direction to a direction of motion.

[0100] In another exemplary aspect, mechanical buffer 512’ can include a spring 732 configured to push moving frame 504 or balance masses 508 in an opposite direction to a direction of motion.

[0101] During a collision event, collision site 736 of mechanical buffers 512 or 512’ can make contact with a surface of moving frame 504 or balance masses 508. In some aspects, collision site 736 can be of a shape and material that would not cause mechanical damage to moving frame 504 or balance masses 508.

[0102] In some aspects, mechanical buffer 512 can operate in an environment comprising air because damper 730 may not work properly in a vacuum environment and a potential leaking of hydraulic fluid may contaminate the environment.

[0103] In some aspects, mechanical buffer 512’ can operate in any environment (e.g., air, vacuum, etc.) because spring 732 does not risk contaminating the environment. Therefore, electromotive braking force 518 as shown in FIG. 5 can be advantageous in a lithographic apparatus because it can provide a damping function in any environment (e.g., air, vacuum, etc.).

[0104] FIG. 8 shows a braking method 800, according to some aspects. In some aspects, at step 802, a current can be applied to one or more actuator coils (e.g., coils 626a-626f shown in FIG. 6A and 6B) to interact with a magnetic field from one or more magnets (e.g., magnets 622 shown in FIG. 6A and 6B) adjacent to the one or more actuator coils.

[0105] In some aspects, at step 804, a force can be generated to move a moving frame (e.g., moving frame 504 shown in FIG. 5) in a lithographic apparatus in a first direction (e.g. first-direction velocity 514 shown in FIG. 5).

[0106] In some aspects, at step 806, the one or more actuator coils can be shorted during a movement of the moving frame in response to an error scenario to prevent collision damage. In some aspects, the one or more actuator coils can be configured to be selectively shorted in real time.

[0107] In some aspects, at step 808, currents can be induced in the one or more actuator coils caused by the magnetic field from the one or more magnets.

[0108] In some aspects, at step 810, a magnetic field can be generated in the one or more actuator coils opposite to the magnetic field from the one or more magnets to produce an electromotive braking force in a second direction (e.g., electromotive braking force 518) opposite to the first direction of the movement of the moving frame.

[0109] In some aspects, at step 812, a velocity of the moving frame can be reduced before the moving frame crashes into one or more mechanical buffers (e.g., mechanical buffers 512 or 512’). In some aspects, a velocity of the one or more actuator coils can be reduced in any environment to reduce the velocity of the moving frame. In some aspects, a velocity of the one or more magnets can be reduced in any environment to reduce the velocity of the moving frame.

[0110] The method steps of FIG. 8 can be performed in any conceivable order and it is not required that all steps be performed. Moreover, the method steps of FIG. 8 described above merely reflect an example of method steps and are not limiting. That is, further method steps and functions are envisaged based aspects described in reference to FIGS. 1A–7B.

[0111] Various embodiments of the present apparatuses, systems and methods are disclosed in the subsequent list of numbered clauses:

1. A lithographic apparatus, comprising:

an illumination system configured to produce a beam of radiation;

a patterning system configured to impart a pattern on the beam, the patterning system comprising a reticle;

a projection system configured to project a patterned beam on a substrate; and

a reticle stage comprising:

a moving frame configured to move with a predetermined amount of kinetic energy;

one or more balance masses adjacent to the moving frame, wherein the one or more balance masses are configured to absorb reaction forces exerted by the moving frame;

one or more actuators configured to move the moving frame and stop movement of the moving frame through braking in response to an error scenario to prevent collision damage, wherein the one or more actuators are configured to be electrically shorted to generate an electromotive braking force opposite to a direction of motion of the moving frame to reduce the kinetic energy of the moving frame and to reduce a braking distance of the moving frame; and

one or more mechanical buffers configured to absorb a remainder of the kinetic energy of the moving frame remaining after the one or more actuators have completed their braking action in response to the error scenario.

2. The lithographic apparatus of clause 1, wherein the one or more actuators are configured to be selectively shorted in real time.

3. The lithographic apparatus of clause 1, wherein the one or more mechanical buffers comprise shock absorbers, dampers, springs, or a combination thereof configured to operate in an environment comprising air.

4. The lithographic apparatus of clause 3, wherein the one or more actuators comprise one or more coils moving at high velocities adjacent to one or more stationary magnets.

5. The lithographic apparatus of clause 1, wherein the one or more mechanical buffers comprise springs such that only the electromotive braking force provides damping in a vacuum environment.

6. The lithographic apparatus of clause 5, wherein the one or more actuators comprise one or more magnets moving at high velocities adjacent to one or more stationary coils.

7. The lithographic apparatus of clause 1, further comprising a capacitor configured to charge the one or more actuators with energy captured from the electromotive braking force.

8. The lithographic apparatus of clause 1, further comprising:

a wafer stage comprising one or more actuators configured to be electrically shorted to generate an electromotive braking force opposite to a direction of motion of the wafer stage on a linear axis to reduce a braking distance of the wafer stage to prevent collision damage.

9. A braking system, comprising:

a moving frame configured to move with a predetermined amount of kinetic energy;

one or more balance masses adjacent to the moving frame, wherein the one or more balance masses are configured to absorb reaction forces exerted by the moving frame;

one or more actuators configured to move the moving frame and stop movement of the moving frame through braking in response to an error scenario to prevent collision damage, wherein the one or more actuators are configured to be electrically shorted to generate an electromotive braking force opposite to a direction of motion of the moving frame to reduce the kinetic energy of the moving frame and to reduce a braking distance of the moving frame; and

one or more mechanical buffers configured to absorb a remainder of the kinetic energy of the moving frame remaining after the one or more actuators have completed their braking action in response to the error scenario.

10. The braking system of clause 9, wherein the one or more actuators are configured to be selectively shorted in real time.

11. The braking system of clause 9, wherein the one or more mechanical buffers comprise shock absorbers, dampers, springs, or a combination thereof configured to operate in an environment comprising air.

12. The braking system of clause 11, wherein the one or more actuators comprise one or more coils moving at high velocities adjacent to one or more stationary magnets.

13. The braking system of clause 9, wherein the one or more mechanical buffers comprise springs such that only the electromotive braking force provides damping in a vacuum environment.

14. The braking system of clause 13, wherein the one or more actuators comprise one or more magnets moving at high velocities adjacent to one or more stationary coils.

15. The braking system of clause 9, further comprising a capacitor configured to charge the one or more actuators with energy captured from the electromotive braking force.

16. The braking system of clause 9, further comprising:

a wafer stage comprising one or more actuators configured to be electrically shorted to generate an electromotive braking force opposite to a direction of motion of the wafer stage on a linear axis to reduce a braking distance of the wafer stage to prevent collision damage.

17. A braking method comprising:

applying a current to one or more actuator coils to interact with a magnetic field from one or more magnets adjacent to the one or more actuator coils;

generating a force to move a moving frame in a lithographic apparatus in a first direction;

shorting the one or more actuator coils during a movement of the moving frame in response to an error scenario to prevent collision damage;

inducing currents in the one or more actuator coils caused by the magnetic field from the one or more magnets;

generating a magnetic field in the one or more actuator coils opposite to the magnetic field from the one or more magnets to produce an electromotive braking force in a second direction, opposite to the first direction of the movement of the moving frame; and

reducing a velocity of the moving frame before the moving frame crashes into one or more mechanical buffers.

18. The braking method of clause 17, wherein the one or more actuator coils are configured to be selectively shorted in real time.

19. The braking method of clause 17, wherein the reducing the velocity of the moving frame comprises reducing a velocity of the one or more actuator coils in any environment.

20. The braking method of clause 17, wherein the reducing the velocity of the moving frame comprises reducing a velocity of the one or more magnets in any environment.

[0112] The terms “radiation,” “beam,” “light,” “illumination,” or the like can be used herein to refer to one or more types of electromagnetic radiation, for example, ultraviolet (UV) radiation (for example, having a wavelength λ of 365, 248, 193, 157 or 126 nm), extreme ultraviolet (EUV or soft X-ray) radiation (for example, having a wavelength in the range of 5-100 nm such as, for example, 13.5 nm), or hard X-ray working at less than 5 nm, as well as particle beams, such as ion beams or electron beams. Generally, radiation having wavelengths between about 400 to about 700 nm is considered visible radiation; radiation having wavelengths between about 780-3000 nm (or larger) is considered IR radiation. UV refers to radiation with wavelengths of approximately 100-400 nm. Within lithography, the term “UV” also applies to the wavelengths that can be produced by a mercury discharge lamp: G-line 436 nm; H-line 405 nm; and/or, I-line 365 nm. Vacuum UV, or VUV (i.e., UV absorbed by gas), refers to radiation having a wavelength of approximately 100-200 nm. Deep UV (DUV) generally refers to radiation having wavelengths ranging from 126 nm to 428 nm, and in some aspects, an excimer laser can generate DUV radiation used within a lithographic apparatus. It should be appreciated that radiation having a wavelength in the range of, for example, 5-20 nm relates to radiation with a certain wavelength band, of which at least part is in the range of 5-20 nm.

[0113] Although some aspects of the present disclosure are described in the context of lithographic apparatuses in the manufacture of ICs, it should be understood that lithographic apparatuses described herein can be used in other applications, for example, in the manufacture of integrated optical systems, guidance and detection patterns for magnetic domain memories, flat-panel displays, LCDs, thin-film magnetic heads, etc. Those skilled in the art will appreciate that, in the context of such alternative applications, any use of the terms “wafer” or “die” herein can be considered as specific examples of the more general terms “substrate” or “target portion”, respectively. A substrate can be processed before or after exposure in, for example, a track unit (a tool that typically applies a layer of resist to a substrate and develops the exposed resist) and/or a metrology unit. Where applicable, aspects disclosed herein can be applied to such and other substrate processing tools. Furthermore, a substrate can be processed more than once, for example in order to create a multi-layer IC, so that the term substrate used herein can also refer to a substrate that already contains multiple processed layers.

[0114] Furthermore, although some aspects of the present disclosure are described in the context of optical lithography, it should be understood that aspects of the present disclosure are not limited to optical lithography. For example, in imprint lithography, a topography in a patterning device defines the pattern created on a substrate. The topography of the patterning device can be pressed into a layer of resist supplied to the substrate whereupon the resist is cured by applying electromagnetic radiation, heat, pressure or a combination thereof. The patterning device is moved out of the resist leaving a pattern in it after the resist is cured.

[0115] It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present specification is to be interpreted by those skilled in relevant art(s) in light of the teachings herein.

[0116] The present disclosure has been described above with the aid of functional building blocks illustrating the implementation of specified functions and relationships thereof. The boundaries of these functional building blocks have been arbitrarily defined herein for the convenience of the description. Alternate boundaries can be defined so long as the specified functions and relationships thereof are appropriately performed. The foregoing description of specific aspects will so fully reveal the general nature of the present disclosure that others can, by applying knowledge within the skill of the art, readily modify and/or adapt for various applications such specific aspects, without undue experimentation and without departing from the general concept of the present disclosure. Therefore, such adaptations and modifications are intended to be within the meaning and range of equivalents of the disclosed aspects, based on the teaching and guidance presented herein.

[0117] It is to be understood that the Detailed Description section, and not the Summary and Abstract sections, is intended to be used to interpret the claims. The Summary and Abstract sections can set forth one or more, but not necessarily all, aspects of the present disclosure as contemplated by the inventor(s), and thus, are not intended to limit the present disclosure and the appended claims in any way. The breadth and scope of the protected subject matter should not be limited by any of the above-described aspects, but should be defined in accordance with the following claims and their equivalents.

CLAIMS

1. A lithographic apparatus, comprising:

an illumination system configured to produce a beam of radiation;

a patterning system configured to impart a pattern on the beam, the patterning system comprising a reticle;

a projection system configured to project a patterned beam on a substrate; and

a reticle stage comprising:

a moving frame configured to move with a predetermined amount of kinetic energy;

one or more balance masses adjacent to the moving frame, wherein the one or more balance masses are configured to absorb reaction forces exerted by the moving frame;

one or more actuators configured to move the moving frame and stop movement of the moving frame through braking in response to an error scenario to prevent collision damage, wherein the one or more actuators are configured to be electrically shorted to generate an electromotive braking force opposite to a direction of motion of the moving frame to reduce the kinetic energy of the moving frame and to reduce a braking distance of the moving frame; and

one or more mechanical buffers configured to absorb a remainder of the kinetic energy of the moving frame remaining after the one or more actuators have completed their braking action in response to the error scenario.

2. The lithographic apparatus of claim 1, wherein the one or more actuators are configured to be selectively shorted in real time.

3. The lithographic apparatus of claim 1, wherein:

the one or more mechanical buffers comprise shock absorbers, dampers, springs, or a combination thereof configured to operate in an environment comprising air; and

the one or more actuators comprise one or more coils moving at high velocities adjacent to one or more stationary magnets.

4. The lithographic apparatus of claim 1, wherein:

the one or more mechanical buffers comprise springs such that only the electromotive braking force provides damping in a vacuum environment; and

the one or more actuators comprise one or more magnets moving at high velocities adjacent to one or more stationary coils.

5. The lithographic apparatus of claim 1, further comprising a capacitor configured to charge the one or more actuators with energy captured from the electromotive braking force.

6. The lithographic apparatus of claim 1, further comprising:

a wafer stage comprising one or more actuators configured to be electrically shorted to generate an electromotive braking force opposite to a direction of motion of the wafer stage on a linear axis to reduce a braking distance of the wafer stage to prevent collision damage.

7. A braking system, comprising:

a moving frame configured to move with a predetermined amount of kinetic energy;

one or more balance masses adjacent to the moving frame, wherein the one or more balance masses are configured to absorb reaction forces exerted by the moving frame;

one or more actuators configured to move the moving frame and stop movement of the moving frame through braking in response to an error scenario to prevent collision damage, wherein the one or more actuators are configured to be electrically shorted to generate an electromotive braking force opposite to a direction of motion of the moving frame to reduce the kinetic energy of the moving frame and to reduce a braking distance of the moving frame; and

one or more mechanical buffers configured to absorb a remainder of the kinetic energy of the moving frame remaining after the one or more actuators have completed their braking action in response to the error scenario.

8. The braking system of claim 7, wherein the one or more actuators are configured to be selectively shorted in real time.

9. The braking system of claim 7, wherein:

the one or more mechanical buffers comprise shock absorbers, dampers, springs, or a combination thereof configured to operate in an environment comprising air; and

the one or more actuators comprise one or more coils moving at high velocities adjacent to one or more stationary magnets.

10. The braking system of claim 7, wherein:

the one or more mechanical buffers comprise springs such that only the electromotive braking force provides damping in a vacuum environment; and

the one or more actuators comprise one or more magnets moving at high velocities adjacent to one or more stationary coils.

11. The braking system of claim 7, further comprising a capacitor configured to charge the one or more actuators with energy captured from the electromotive braking force.

12. The braking system of claim 7, further comprising:

a wafer stage comprising one or more actuators configured to be electrically shorted to generate an electromotive braking force opposite to a direction of motion of the wafer stage on a linear axis to reduce a braking distance of the wafer stage to prevent collision damage.

13. A braking method comprising:

applying a current to one or more actuator coils to interact with a magnetic field from one or more magnets adjacent to the one or more actuator coils;

generating a force to move a moving frame in a lithographic apparatus in a first direction;

shorting the one or more actuator coils during a movement of the moving frame in response to an error scenario to prevent collision damage;

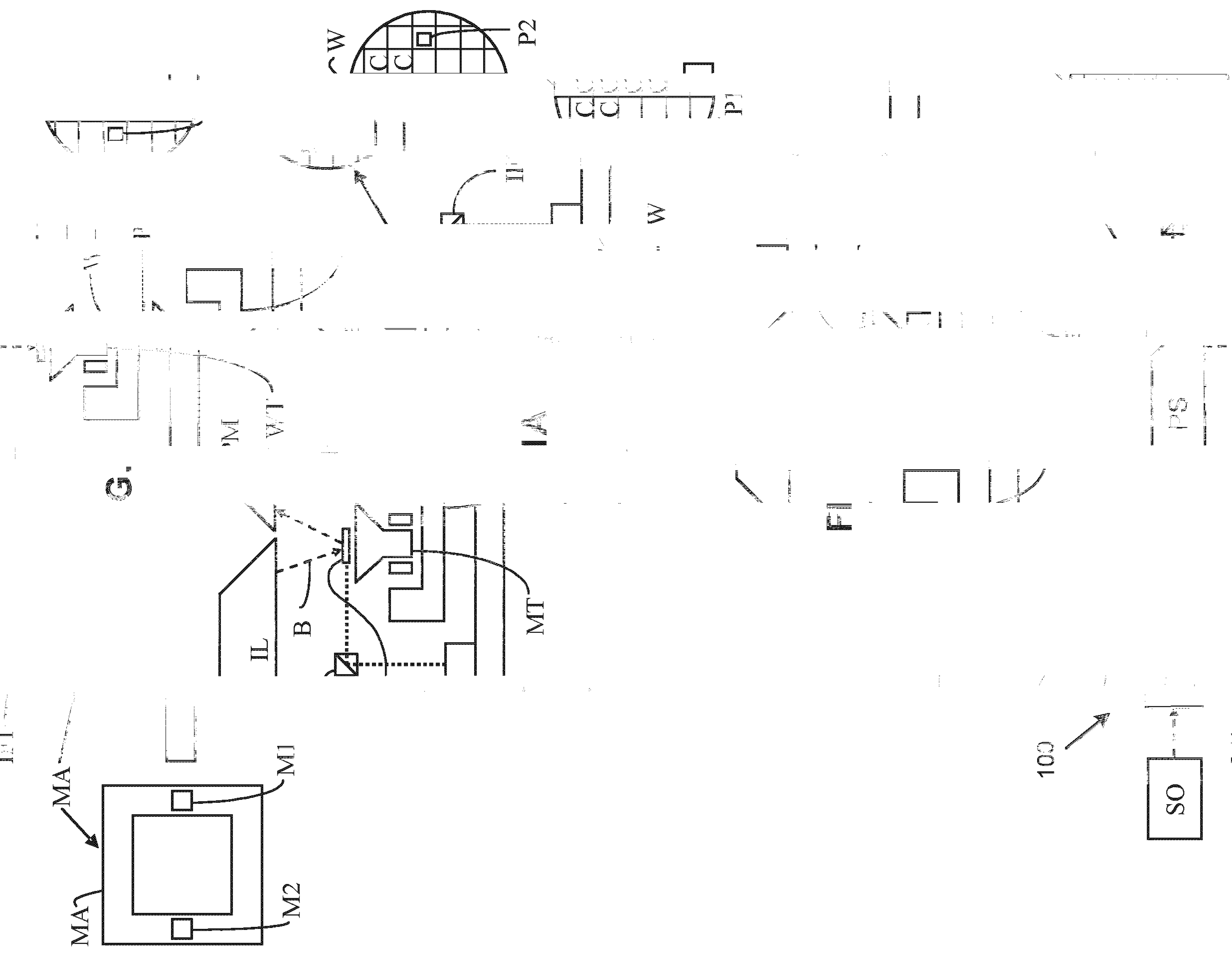
inducing currents in the one or more actuator coils caused by the magnetic field from the one or more magnets;

generating a magnetic field in the one or more actuator coils opposite to the magnetic field from the one or more magnets to produce an electromotive braking force in a second direction, opposite to the first direction of the movement of the moving frame; and

reducing a velocity of the moving frame before the moving frame crashes into one or more mechanical buffers.

14. The braking method of claim 13, wherein the one or more actuator coils are configured to be selectively shorted in real time.

15. The braking method of claim 13, wherein the reducing the velocity of the moving frame comprises reducing a velocity of the one or more actuator coils in any environment or reducing a velocity of the one or more magnets in any environment.



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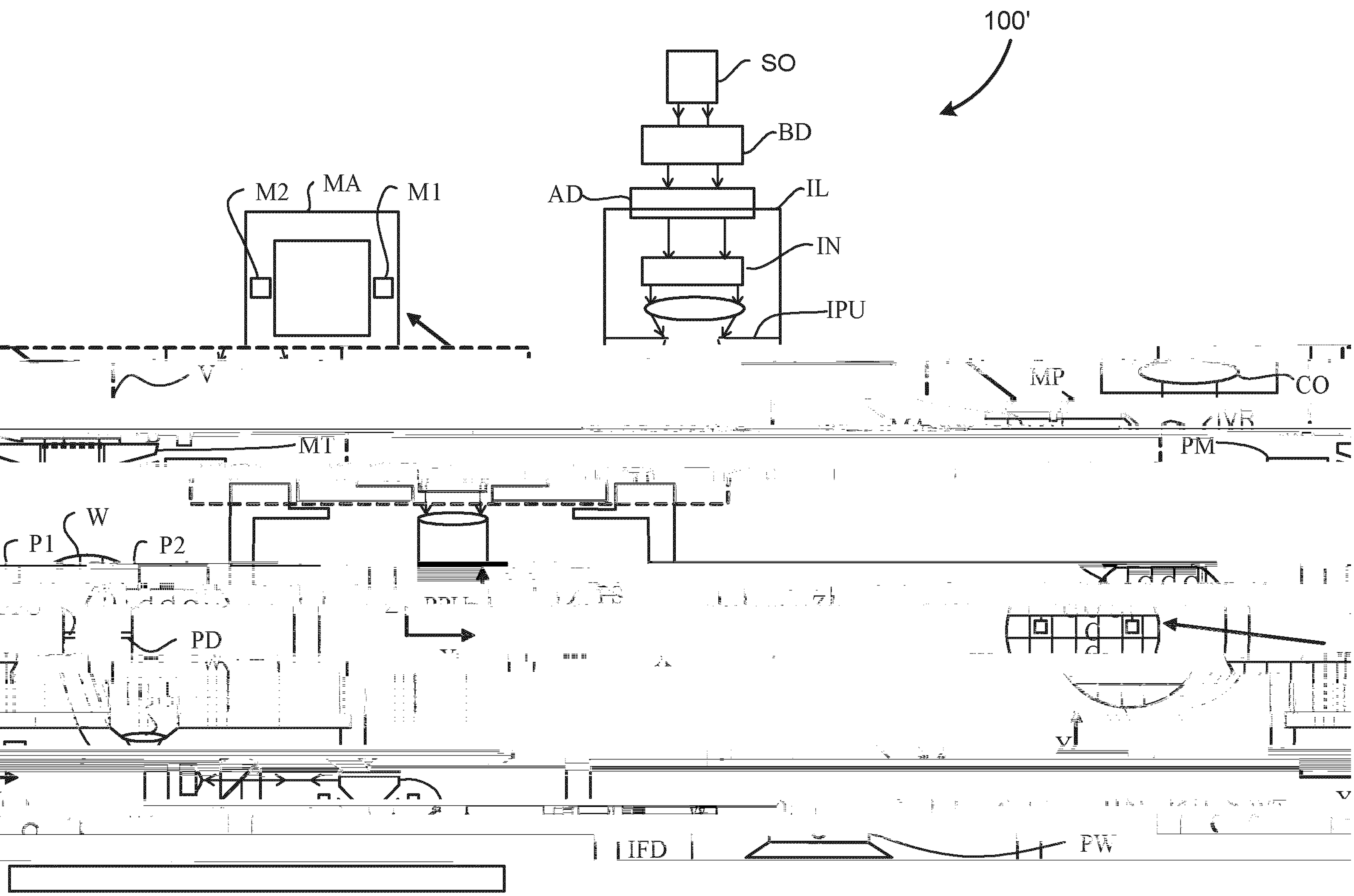
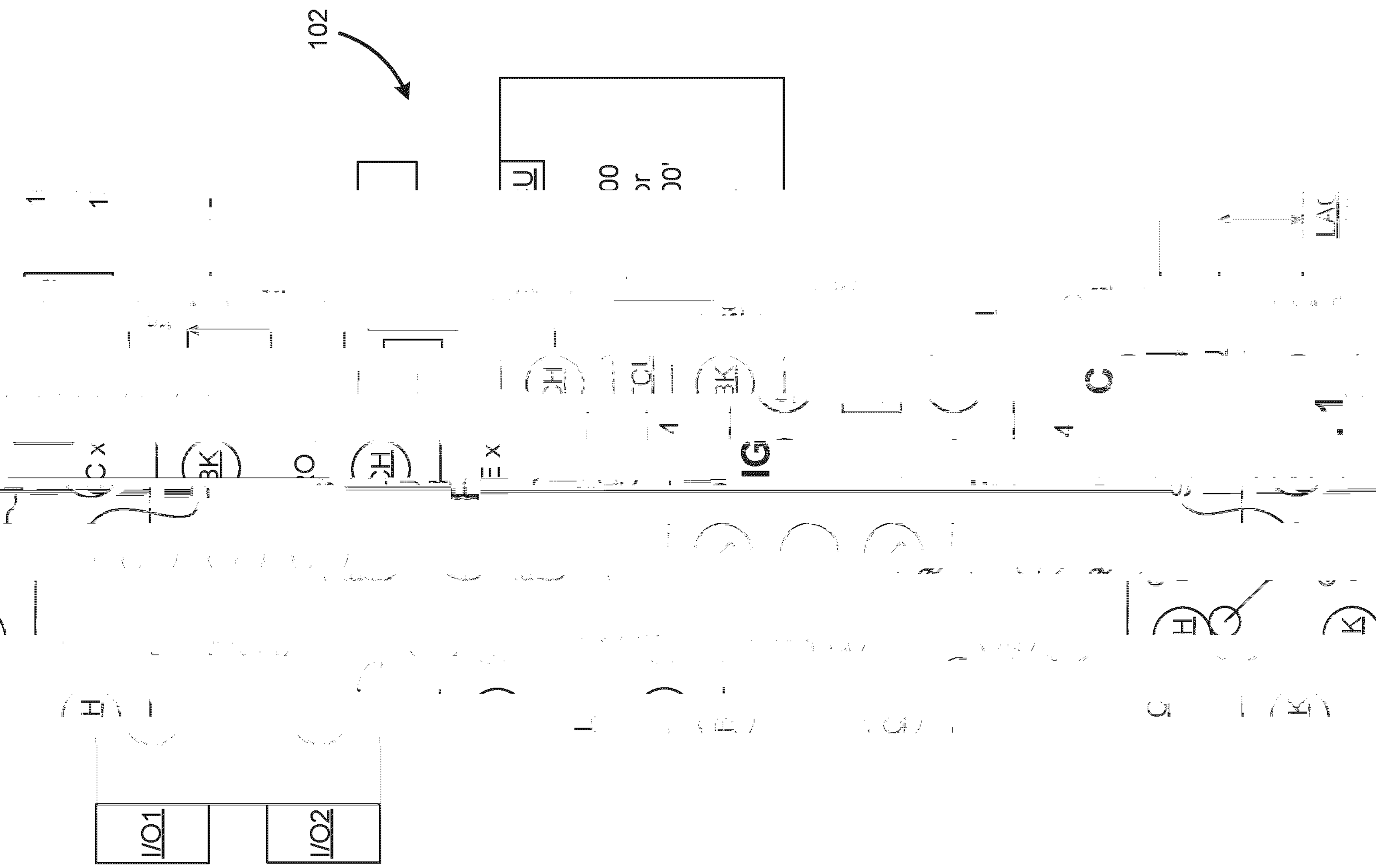


FIG. 1B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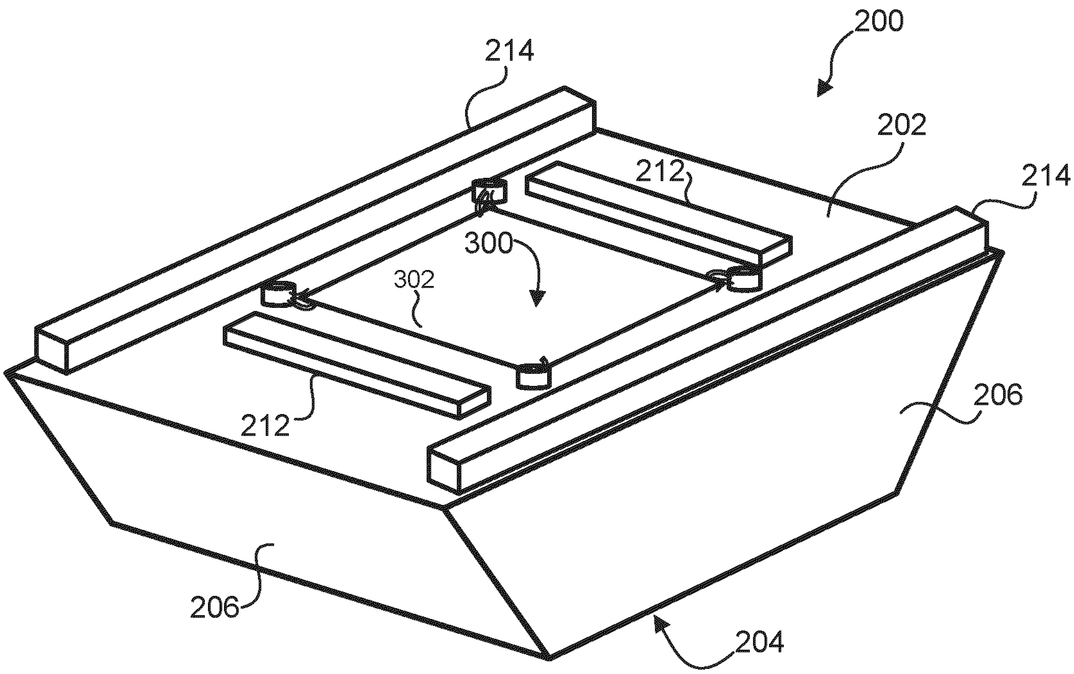
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FIG. 2

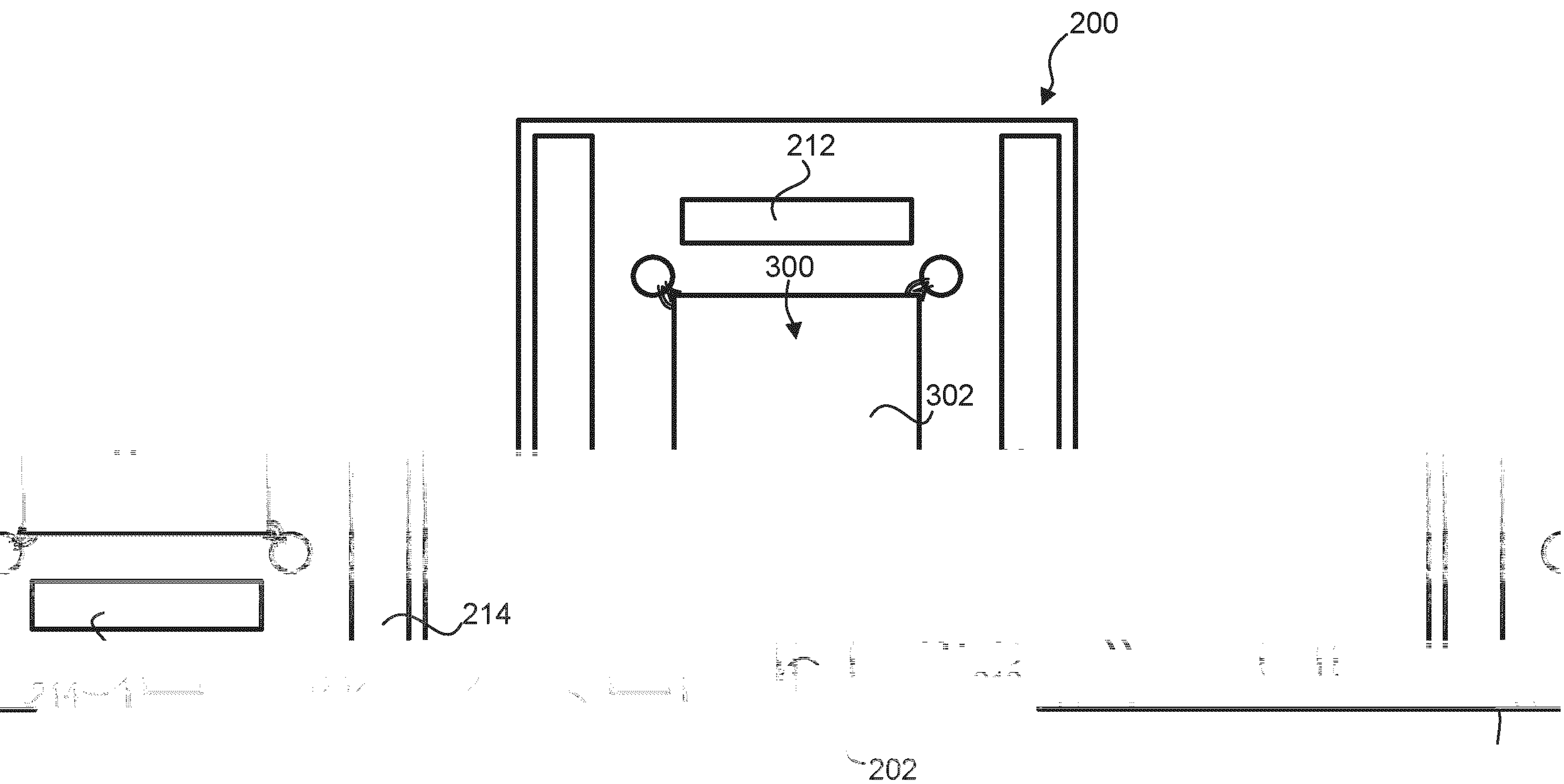
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FIG. 3

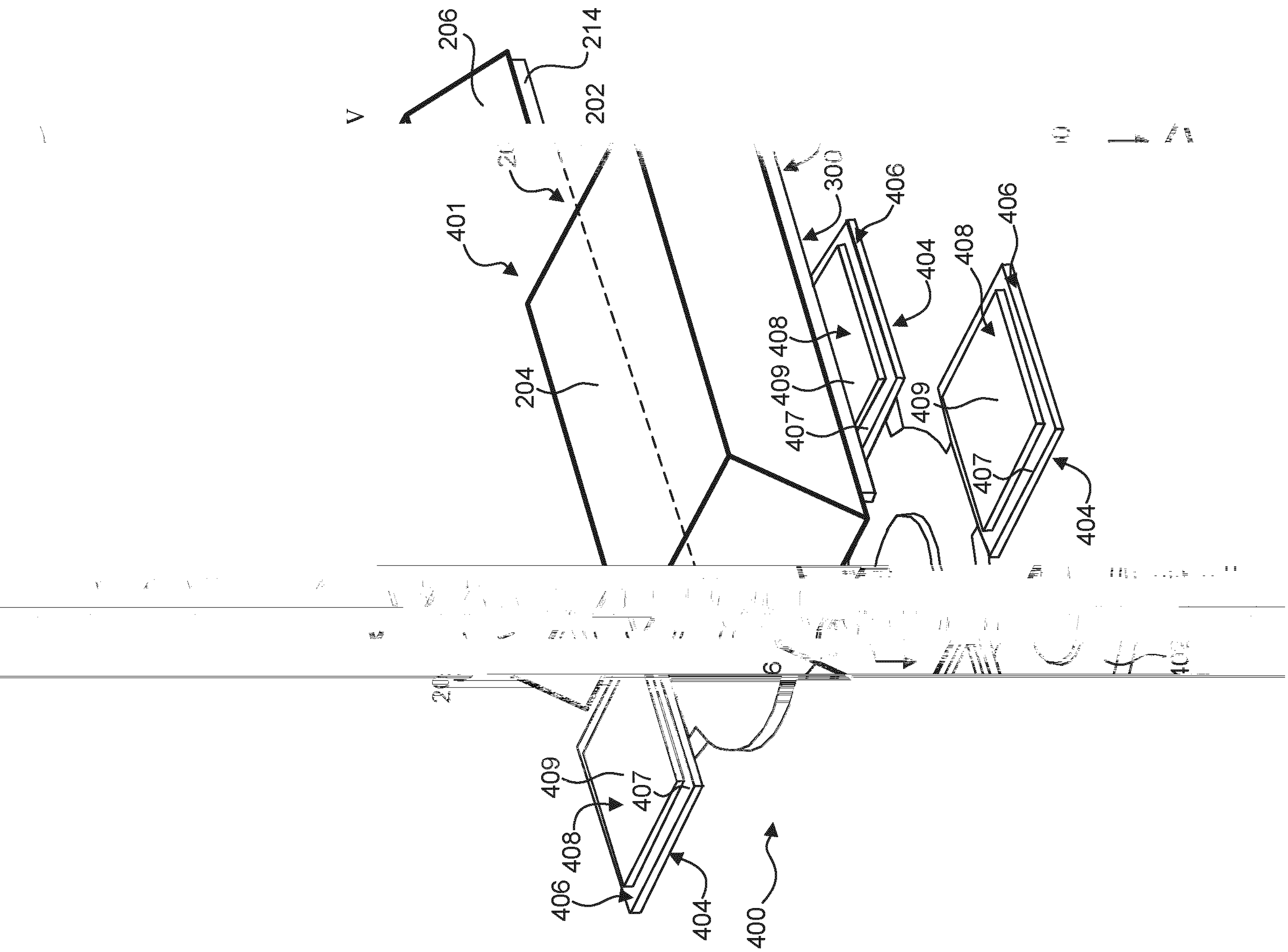

FIG. 4

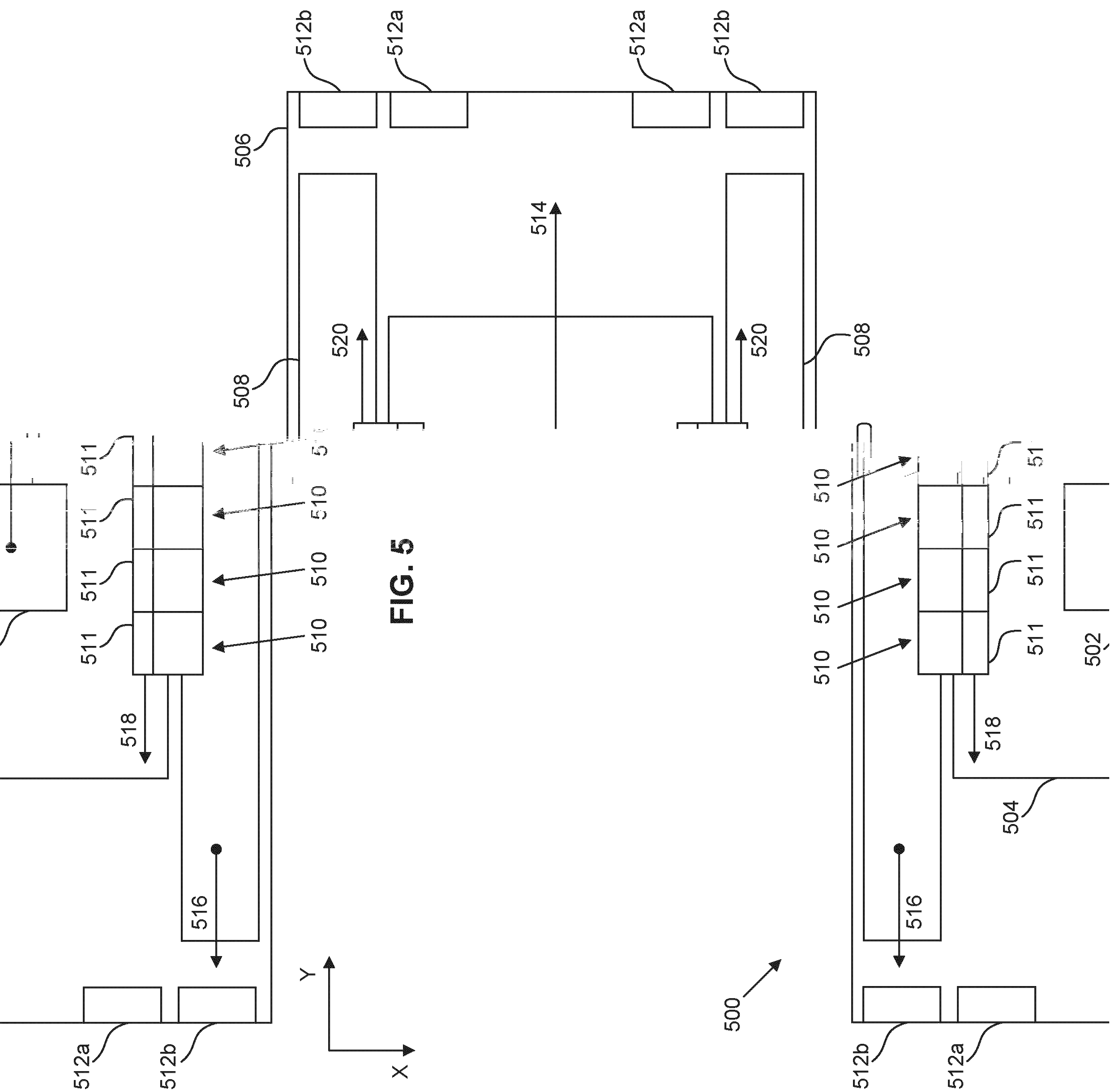
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FIG. 5

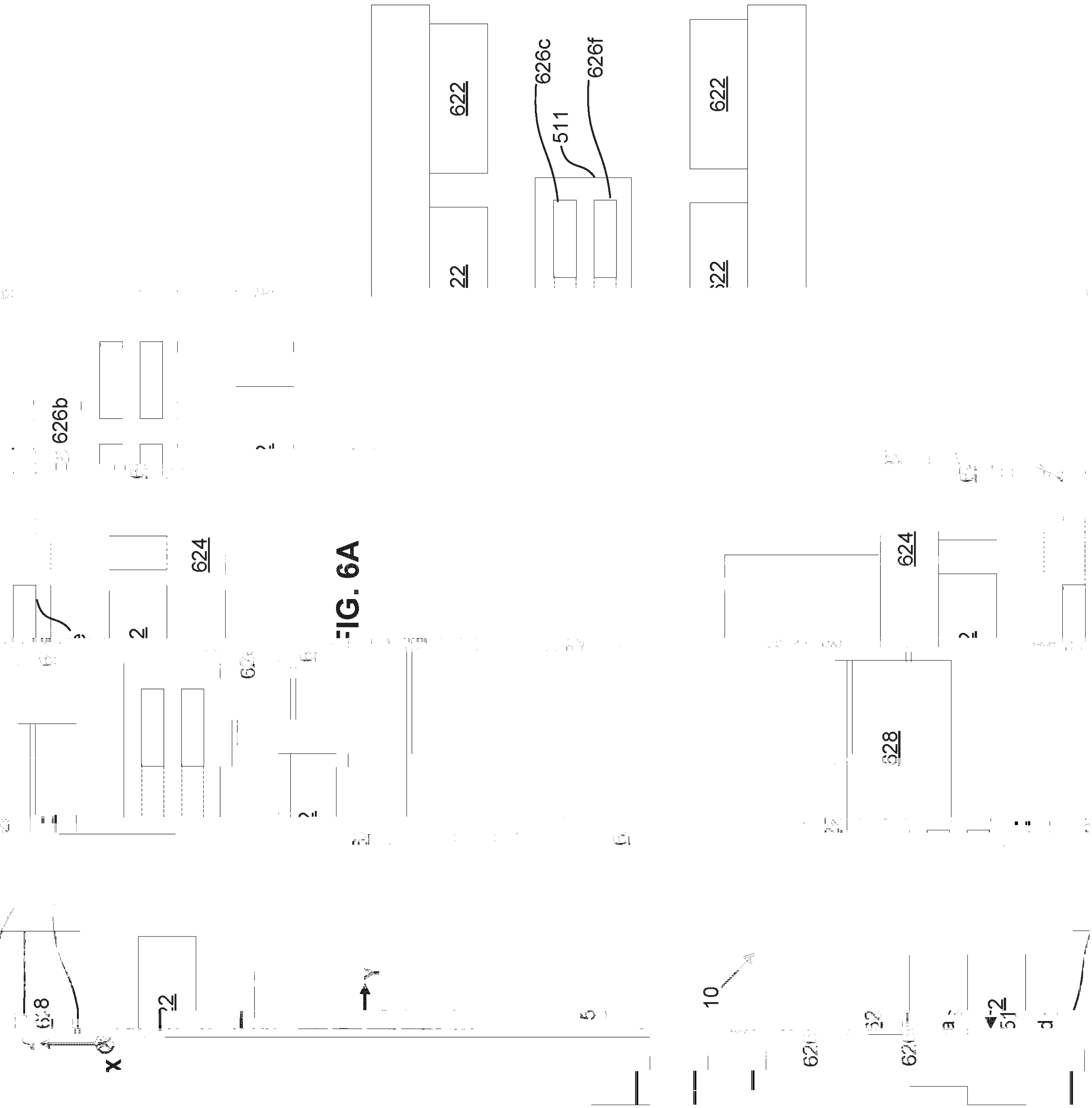

FIG. 6A

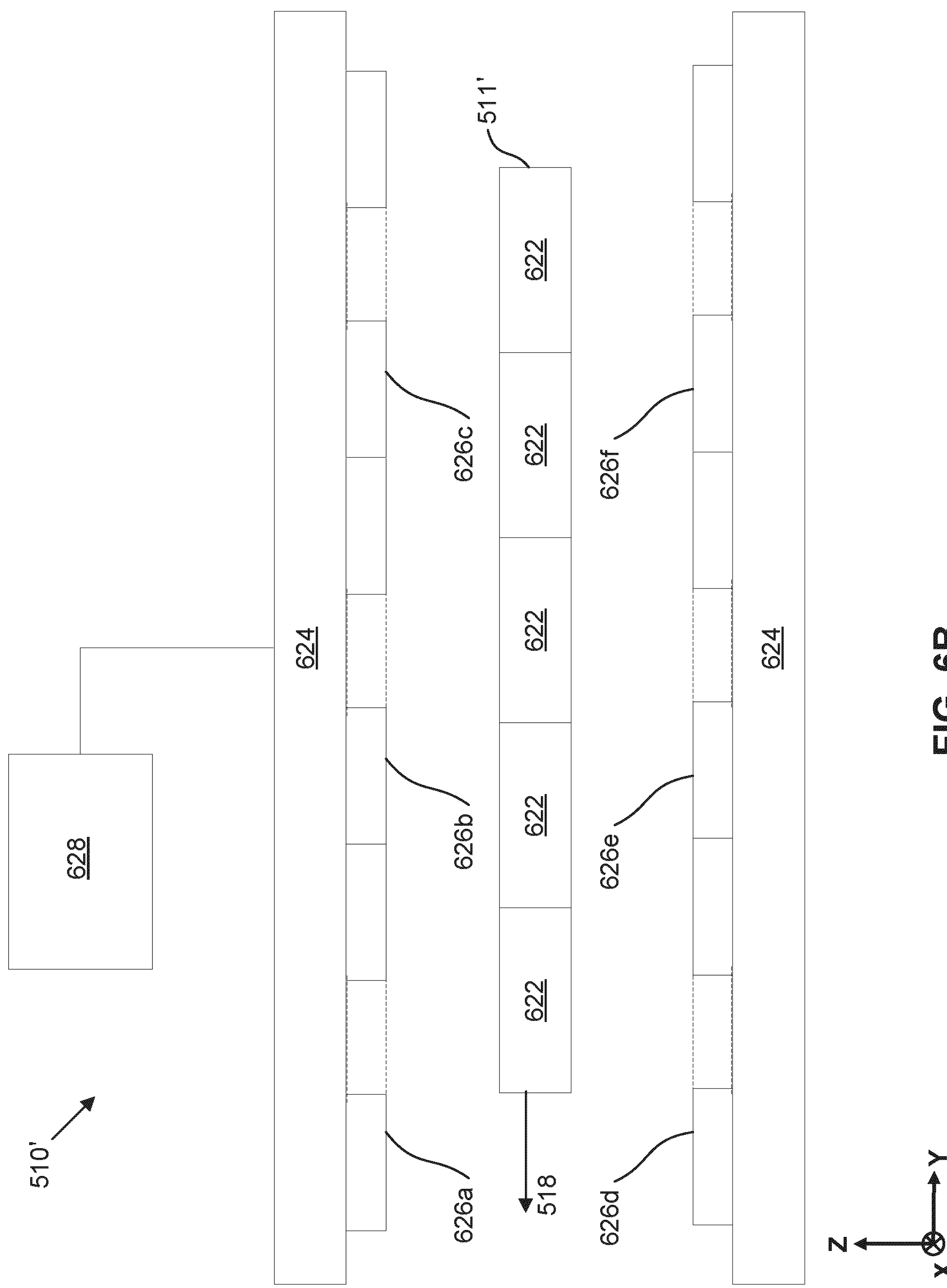
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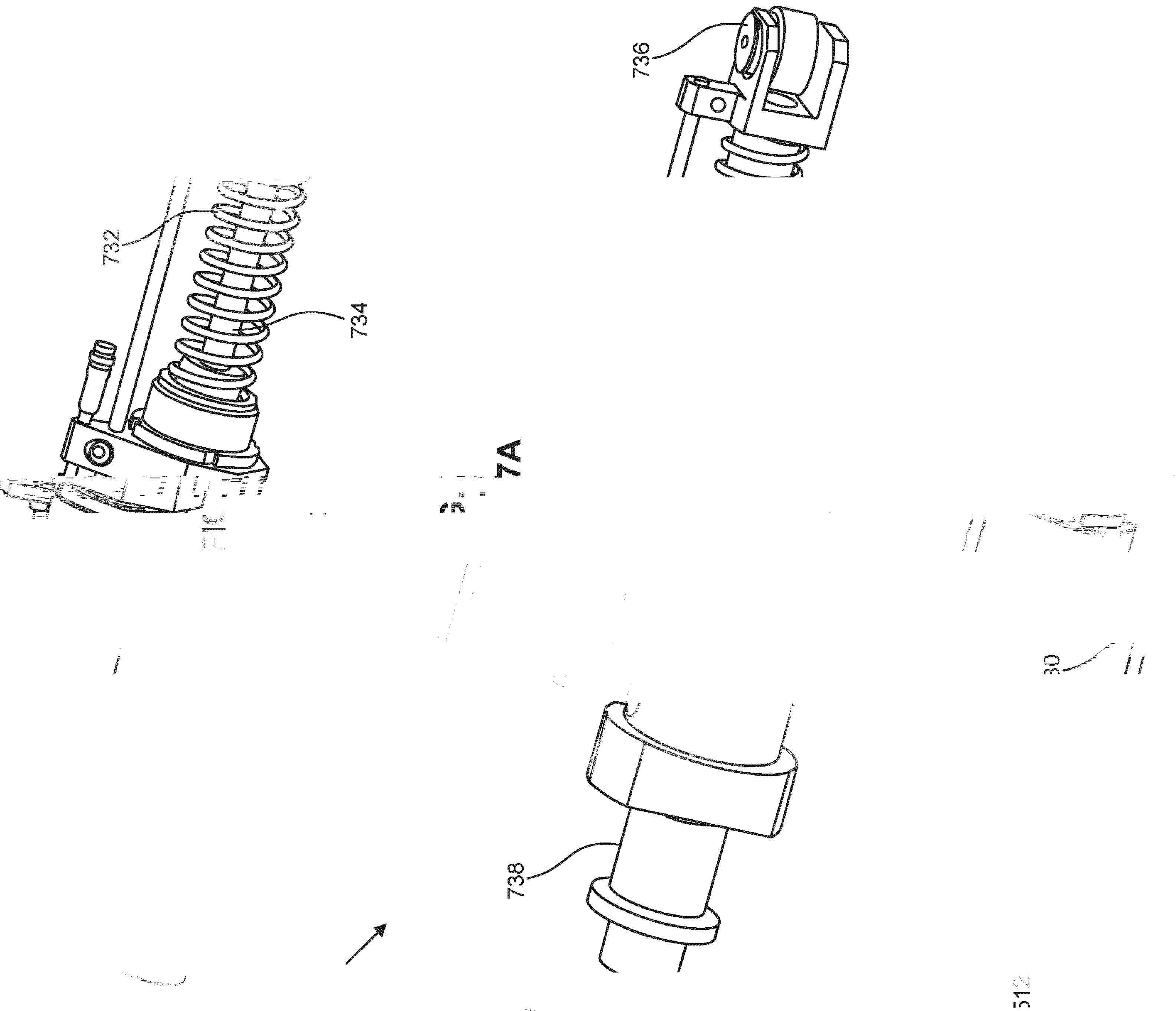
FIG. 6B



7A

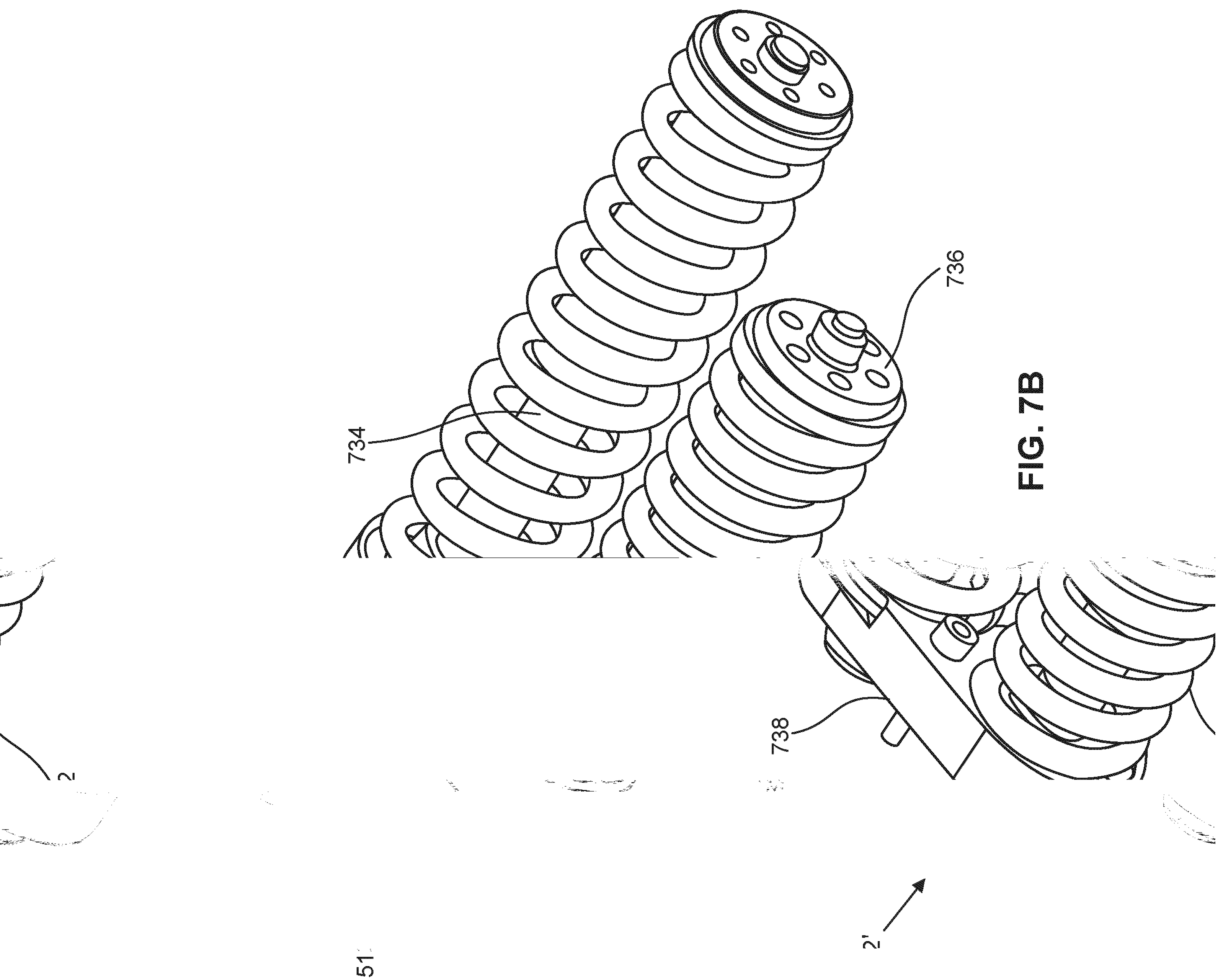


FIG. 7B

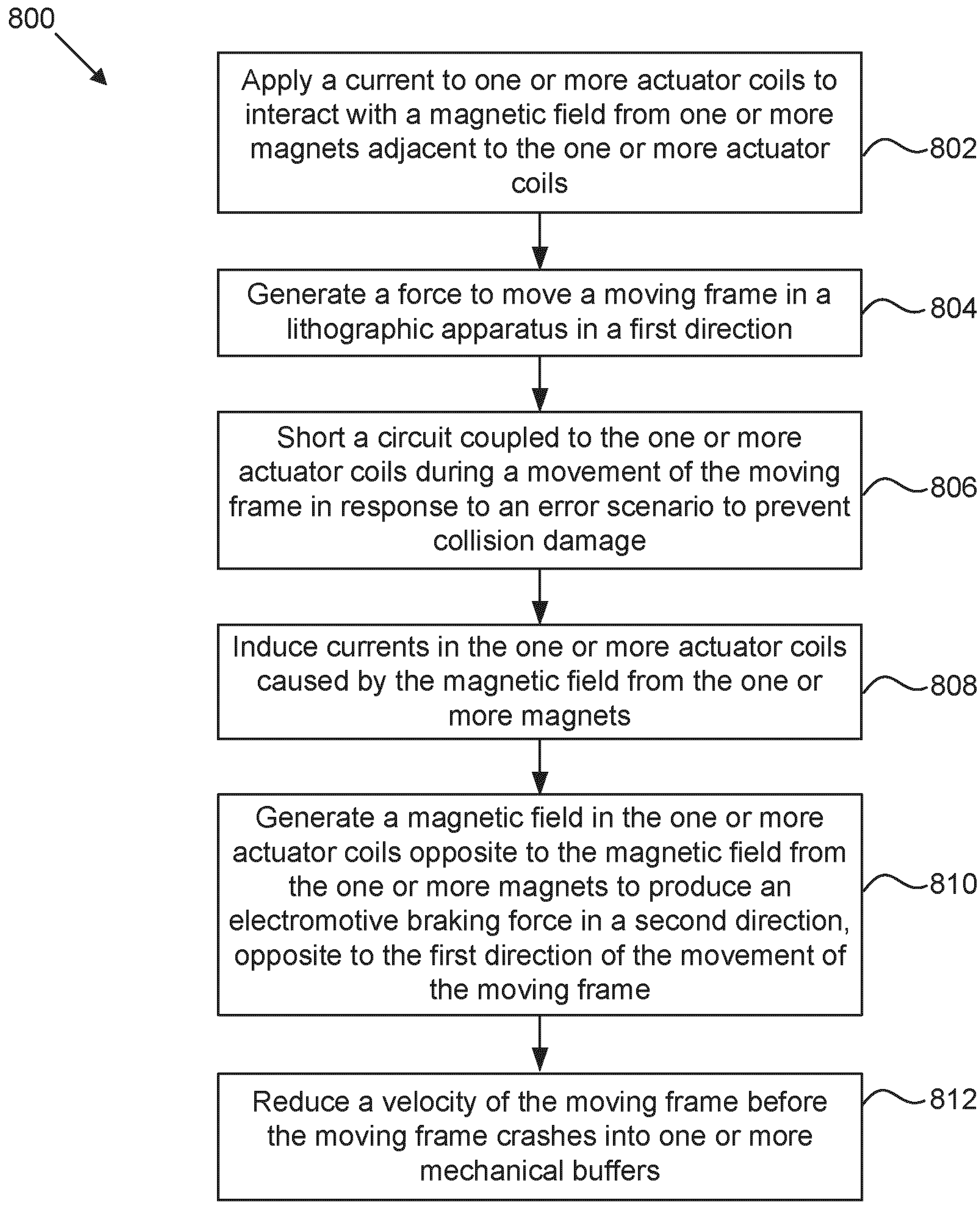

800
Apply a current to one or more actuator coils to interact with a magnetic field from one or more magnets adjacent to the one or more actuator coils
802
Generate a force to move a moving frame in a lithographic apparatus in a first direction
804
Short a circuit coupled to the one or more actuator coils during a movement of the moving frame in response to an error scenario to prevent collision damage
806
Induce currents in the one or more actuator coils caused by the magnetic field from the one or more magnets
808
Generate a magnetic field in the one or more actuator coils opposite to the magnetic field from the one or more magnets to produce an electromotive braking force in a second direction, opposite to the first direction of the movement of the moving frame
810
Reduce a velocity of the moving frame before the moving frame crashes into one or more mechanical buffers
812

FIG. 8

INTERNATIONAL SEARCH REPORT

International application No
PCT/EP2024/071837

A. CLASSIFICATION OF SUBJECT MATTER
INV. G03F7/00
ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)
G03F

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

EPO-Internal, WPI Data

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
Y	WO 2018/050443 A1 (ASML NETHERLANDS BV [NL]; ASML HOLDING NV [NL]) 22 March 2018 (2018-03-22) paragraph [0019] - paragraph [0055] figures 1-3 -----	1-15
Y	US 2008/170213 A1 (OHISHI SHINJI [JP]) 17 July 2008 (2008-07-17) paragraph [0009] - paragraph [0062] figure 8 -----	1-15
Y	WO 2007/017802 A2 (KONINKL PHILIPS ELECTRONICS NV [NL]; HOOGZAAD GIAN [NL]) 15 February 2007 (2007-02-15)	5,11
A	page 4, line 11 - line 15 -----	1,7,13

☐ Further documents are listed in the continuation of Box C.

☒ See patent family annex.

* Special categories of cited documents :

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"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance;; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance;; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
14 November 2024	26/11/2024

Name and mailing address of the ISA/	Authorized officer
European Patent Office, P.B. 5818 Patentlaan 2 NL - 2280 HV Rijswijk Tel. (+31-70) 340-2040, Fax: (+31-70) 340-3016	Meixner, Matthias

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No
PCT/EP2024/071837

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